

ASVH-ARCH-001

Runtime Architecture

**Independent Runtime Authority Architecture for AI-enabled Ambient
Scribing**

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Engineering Reference Architecture

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- ASVH-METH-001
- ASVH-SPEC-001
- ASVH-TRACE-001
- ASVH-001 (Reference Implementation Report)

Executive Summary

Artificial Intelligence is rapidly transforming the creation of clinical documentation within healthcare. AI-enabled ambient scribing products are now being deployed across NHS organisations to assist clinicians by capturing consultations and generating draft clinical records. Recognising both the opportunities and risks associated with these technologies, NHS England has published guidance defining the governance, clinical safety and organisational responsibilities necessary for their safe deployment.

While this guidance establishes the conditions under which ambient scribing systems may be introduced into clinical practice, it does not prescribe an implementation-independent architectural mechanism for determining whether AI-generated documentation should still be permitted to create an institutional consequence immediately before execution. Existing ambient scribing architectures primarily focus on consultation capture, AI-assisted documentation generation, clinician review and Electronic Patient Record (EPR) integration. They do not define an independent runtime capability responsible for evaluating whether delegated authority remains legitimately exercisable at the point where an approved action becomes an institutional record.

The **Ambient Scribing Verification Harness (ASVH)** addresses this architectural gap.

ASVH defines an implementation-independent **Runtime Architecture** centred upon an **Independent Runtime Authority** positioned at the **Execution Bind Point** immediately before AI-generated documentation is committed to the Electronic Patient Record. Rather than replacing governance, clinical safety processes, clinician judgement or ambient scribing platforms, the Runtime Authority introduces an additional architectural responsibility: the deterministic evaluation of runtime admissibility using the complete operational context that exists immediately before institutional consequence formation.

The architecture is intentionally implementation independent. It does not prescribe a specific ambient scribing supplier, Electronic Patient Record, deployment model or technology platform. Instead, it defines the architectural principles, responsibilities and interactions required to introduce deterministic runtime verification into existing healthcare workflows while preserving clear separation between governance, AI generation, clinical accountability and execution.

This document defines:

- the architectural principles governing Runtime Authority;
- the architectural components participating in runtime verification;
- the Runtime Context evaluated immediately before execution;
- the Runtime Evaluation Sequence performed at the Execution Bind Point;
- the Runtime Decision Architecture used to produce deterministic execution outcomes;
- the Authority Receipt Architecture providing immutable engineering evidence; and
- implementation-independent deployment principles supporting integration with existing healthcare environments.

The architecture is supported by a wider engineering reference programme comprising **ASVH-METH-001** (Runtime Requirements Engineering Methodology), **ASVH-TRACE-001** (Guidance Traceability Matrix), **ASVH-SPEC-001** (Runtime Authority Specification) and **ASVH-001** (Reference Implementation). Together, these publications provide a complete engineering pathway from published NHS guidance through to deterministic runtime verification and reference implementation.

The ASVH Runtime Architecture does **not** define clinical practice, AI model behaviour, governance policy, supplier implementation or Electronic Patient Record design. Those responsibilities remain with NHS organisations, ambient scribing suppliers, clinicians and existing healthcare governance frameworks. Instead, ASVH defines a complementary architectural capability that determines whether an approved action remains admissible immediately before execution, using deterministic evaluation of the current Runtime Context.

The principal contribution of this document is the introduction of an **Independent Runtime Authority** as a distinct architectural layer within AI-enabled ambient scribing workflows. By separating runtime admissibility from governance, AI generation and clinical judgement, the architecture provides an implementation-independent reference model that enables deterministic execution-time verification while preserving evidential accountability, architectural independence and clear separation of responsibility. This reference architecture is intended to support healthcare organisations, ambient scribing suppliers and enterprise architects exploring the integration of runtime verification into future AI-enabled clinical systems.

Architecture at a Glance

What problem does ASVH solve?

The NHS England guidance for AI-enabled ambient scribing defines how organisations should govern, deploy and clinically assure AI-assisted documentation systems. It also establishes the responsibilities of suppliers, NHS organisations and clinicians throughout the deployment lifecycle.

What the guidance does not define is an architectural mechanism for determining whether AI-generated documentation should still be permitted to create an institutional consequence **at the precise moment execution occurs**.

ASVH addresses that architectural gap.

Rather than introducing another governance framework, ASVH defines an **implementation-independent Runtime Authority** that performs deterministic execution-time verification immediately before AI-generated documentation is committed to the Electronic Patient Record (EPR).

The Architectural Contribution

The ASVH Runtime Architecture introduces one new architectural responsibility:

The independent determination of whether delegated authority remains legitimately exercisable immediately before institutional consequence formation.

Everything else already exists:

- NHS Governance
- Clinical Safety
- Ambient Scribing Platforms
- Responsible Clinicians
- Electronic Patient Records

ASVH introduces only the missing runtime decision layer.

What ASVH Does Not Replace

ASVH has been designed to complement existing healthcare architecture rather than replace it.

It does **not** replace:

- NHS England guidance
- DCB0129 clinical safety activities
- DCB0160 deployment responsibilities
- Responsible clinician judgement
- Ambient scribing products
- Electronic Patient Record systems

Instead, Runtime Authority evaluates whether execution remains admissible using the operational state that exists immediately before institutional consequence formation.

Architectural Principles

The architecture is founded upon six implementation-independent principles.

Principle	Description
Independence	Runtime Authority evaluates execution without assuming responsibility for governance, AI generation or clinical judgement.
Determinism	Identical Runtime Contexts always produce identical Runtime Decisions.
Execution-Time Verification	Evaluation occurs immediately before institutional consequence formation.
Implementation Independence	Architectural behaviour remains independent of vendors and deployment technologies.

Separation of Responsibility	Governance, execution, runtime evaluation and evidence generation remain distinct architectural concerns.
Evidential Accountability	Every Runtime Decision is supported by an immutable Authority Receipt.

The ASVH Engineering Framework

This document forms one component of a broader engineering reference project.

Engineering Activity	Supporting Publication
Runtime Requirements Engineering	ASVH-METH-001
Guidance Traceability	ASVH-TRACE-001
Runtime Rule Specification	ASVH-SPEC-001
Runtime Architecture	ASVH-ARCH-001
Reference Implementation	ASVH-001

Together these publications define the complete engineering pathway from published NHS guidance through to deterministic runtime verification.

Scope of this Document

This publication defines the **implementation-independent reference architecture** supporting Runtime Authority within AI-enabled ambient scribing workflows.

It defines:

- architectural principles;
- architectural components;
- Runtime Context;
- Runtime Decision Architecture;
- Authority Receipt Architecture;
- implementation-independent deployment principles.

It does **not** define runtime rules, implementation technologies or software behaviour, which are specified in the companion ASVH publications.

1. Purpose

Artificial Intelligence is increasingly being used to assist the creation of clinical documentation within healthcare environments. NHS England has published guidance describing the governance, clinical safety and organisational responsibilities associated with the deployment of AI-enabled ambient scribing technologies.

While these governance activities establish the conditions under which ambient scribing may be safely deployed, they do not prescribe how software should determine, at the point of execution, whether AI-generated clinical documentation should still be committed to the EPR (EPR).

This document defines the reference runtime architecture developed for the **ASVH-001 – NHS Ambient Scribing Runtime Verification Harness**.

The architecture demonstrates how an **Independent Runtime Authority** may be positioned at the **Execution Bind Point** to evaluate runtime admissibility immediately before institutional consequence formation.

The architecture is deliberately **implementation independent**. It does not prescribe the internal design of ambient scribing products, EPR systems or healthcare infrastructure. Instead, it provides a reference architectural model that illustrates how deterministic runtime verification may complement existing governance, clinical safety processes and clinical accountability.

The purpose of this document is to define:

- the architectural components participating in runtime verification;
- the responsibilities assigned to each architectural component;
- the execution boundaries between AI generation, clinical approval and Runtime Authority;
- the Runtime Context evaluated immediately before execution;
- the architectural placement of the Execution Bind Point;
- the generation of Authority Receipts as deterministic execution evidence.

Together with **ASVH-METH-001**, **ASVH-SPEC-001** and **ASVH-TRACE-001**, this document completes the architectural foundation of the ASVH engineering reference project and provides the blueprint for implementation-independent Runtime Authority within AI-enabled ambient scribing workflows.

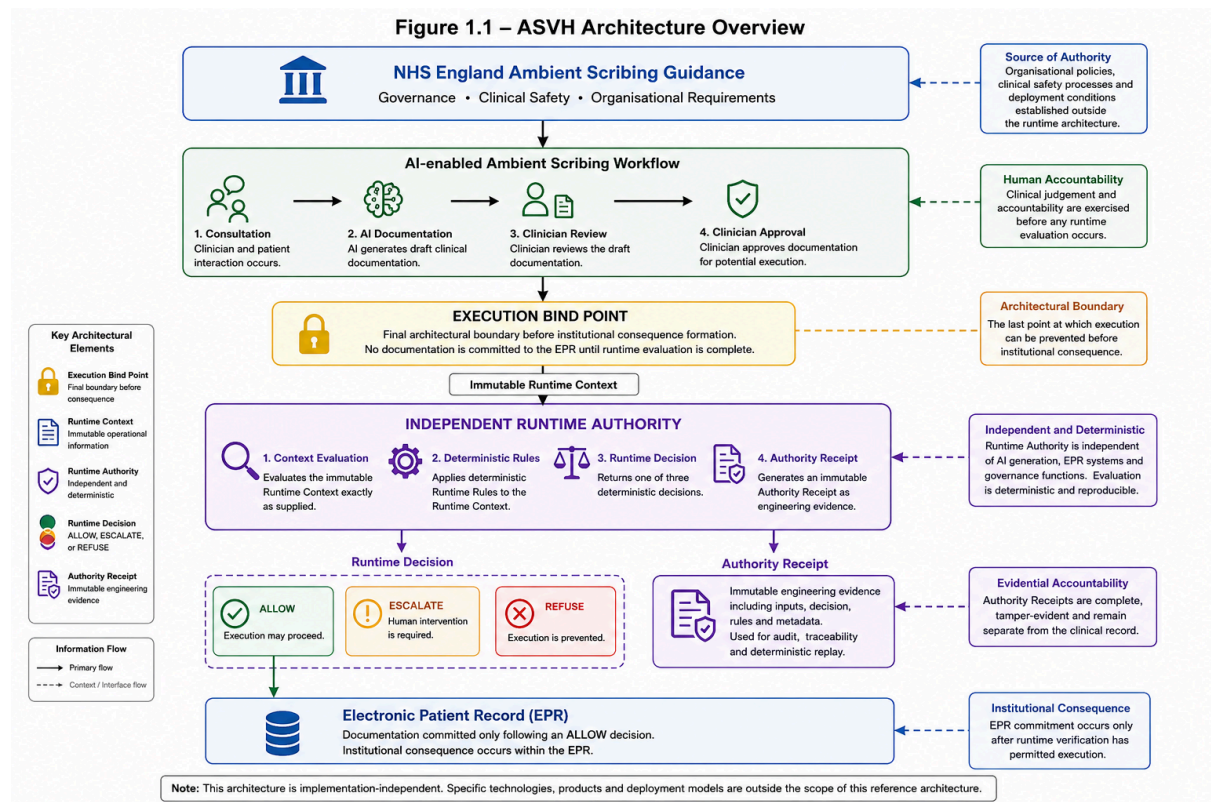


Figure 1.1 – ASVH Architecture Overview

Description

Figure 1.1 provides a high-level overview of the ASVH Runtime Architecture and illustrates the overall execution model before the detailed architectural chapters are introduced.

The figure shows that the architecture begins with the organisational operating conditions established through NHS England guidance and local governance. AI-enabled ambient scribing systems generate draft clinical documentation that is reviewed and approved by the responsible clinician before any institutional execution is permitted.

Following clinician approval, execution is intentionally suspended at the **Execution Bind Point**, representing the final architectural boundary before institutional consequence formation. At this point an immutable **Runtime Context** is assembled and submitted to an independent **Runtime Authority** for deterministic evaluation.

The Runtime Authority evaluates the supplied Runtime Context using deterministic Runtime Rules and returns one of three execution decisions: **ALLOW**, **ESCALATE** or **REFUSE**. At the same time, an immutable **Authority Receipt** is generated to provide engineering evidence supporting the runtime determination.

Only an **ALLOW** decision permits AI-generated documentation to be committed to the **Electronic Patient Record (EPR)**, where institutional consequence occurs. The Authority

Receipt remains separate from the clinical record and provides independent evidence supporting audit, traceability and deterministic replay.

This overview introduces the principal architectural concepts developed throughout the remainder of this document, including the Execution Bind Point, Runtime Context, Runtime Authority, Runtime Decision and Authority Receipt, and provides readers with a conceptual roadmap for the detailed architecture presented in subsequent chapters.

2. Scope

The ASVH Runtime Architecture defines an implementation-independent reference architecture for evaluating runtime admissibility immediately before AI-generated clinical documentation is committed to the EPR (EPR).

The architecture describes the components, interfaces and execution boundaries required to support deterministic Runtime Authority evaluation within an AI-enabled ambient scribing workflow.

The scope of this document is intentionally limited to runtime execution. It does not attempt to define governance policy, clinical practice or supplier implementation. Those responsibilities remain with the appropriate organisational and technical stakeholders.

2.1 In Scope

This document defines:

- the architectural components participating in runtime verification;
- the responsibilities assigned to each architectural component;
- the Runtime Context evaluated immediately before execution;
- the placement and purpose of the Execution Bind Point;
- the logical interaction between the Runtime Authority and the surrounding workflow;
- deterministic runtime rule evaluation;
- Authority Receipt generation;
- architectural interfaces between runtime verification and downstream execution;
- implementation-independent reference deployment principles.

2.2 Out of Scope

The following responsibilities are explicitly outside the scope of the ASVH Runtime Architecture:

Clinical Practice

- diagnosis;
- treatment planning;
- clinical reasoning;
- documentation quality;
- clinical decision making.

Ambient Scribing Products

- speech recognition;
- transcription;
- language model execution;

- prompt engineering;
- clinical summarisation.

EPR Systems

- record storage;
- data persistence;
- EPR workflows;
- clinical data models;
- vendor-specific implementation.

Governance

- procurement;
- supplier assurance;
- organisational policy;
- DCB0129 and DCB0160 compliance activities;
- clinical safety case development;
- deployment approval.

Infrastructure

- cloud architecture;
- networking;
- authentication;
- identity management;
- database implementation;
- hosting environments.

2.3 Architectural Boundary

The architectural boundary defined by ASVH begins when an authorised execution request reaches the Execution Bind Point and concludes when a deterministic runtime decision has been returned.

Activities occurring before this boundary—including AI generation, clinician review and governance processes—and activities occurring after this boundary—including EPR persistence and downstream workflow execution—remain outside the responsibilities of the Runtime Authority.

This deliberate separation of responsibility ensures that the Runtime Authority remains focused on a single architectural concern:

Determining whether delegated authority remains legitimately exercisable immediately before institutional consequence formation.

3. Architectural Principles

The ASVH Runtime Architecture is founded upon a small number of architectural principles that govern the separation of responsibility between AI generation, clinical accountability, governance and Runtime Authority.

These principles define the characteristics of the architecture independently of any specific software implementation, ambient scribing supplier or EPR.

3.1 Independent Runtime Evaluation

Runtime Authority shall evaluate execution independently of AI generation.

Ambient scribing platforms are responsible for producing clinical documentation.

Responsible clinicians remain accountable for reviewing and approving that documentation.

The Runtime Authority performs neither function.

Instead, it independently determines whether delegated authority remains legitimately exercisable immediately before institutional consequence formation.

This separation ensures that runtime admissibility is evaluated without influencing the generation of clinical content or the exercise of professional clinical judgement.

3.2 Deterministic Execution

Given an identical Runtime Context, the Runtime Authority shall always produce an identical decision.

No probabilistic reasoning, adaptive learning or autonomous optimisation shall influence runtime evaluation.

Deterministic behaviour ensures that every execution decision can be reproduced, explained and independently verified.

3.3 Separation of Responsibility

The architecture deliberately separates four independent responsibilities:

- **Governance** establishes organisational policy, clinical safety and deployment requirements.
- **Ambient Scribing Products** generate clinical documentation.
- **Responsible Clinicians** review and approve clinical documentation.
- **Runtime Authority** determines whether execution remains admissible immediately before consequence formation.

No architectural component assumes responsibilities assigned to another.

3.4 Minimal Runtime Context

The Runtime Authority evaluates only the information required to determine runtime admissibility.

It does not require access to:

- raw consultation audio;
- language model internals;
- prompt history;
- clinical reasoning;
- governance documentation.

Only the operational state immediately preceding execution is evaluated.

This minimises architectural coupling while supporting implementation independence.

3.5 Implementation Independence

The Runtime Architecture does not depend upon:

- a specific ambient scribing supplier;
- a particular EPR;
- a cloud platform;
- a programming language;
- a workflow engine.

The architecture defines logical responsibilities and interactions rather than prescribing implementation technologies.

3.6 Evidential Accountability

Every runtime evaluation shall generate sufficient evidence to support:

- explanation;
- audit;
- deterministic replay;
- traceability;
- independent verification.

This evidence is recorded within an Authority Receipt.

The Authority Receipt forms an engineering evidence artefact and does not constitute part of the clinical record.

3.7 Execution-Time Determination

The Runtime Authority evaluates admissibility only at the Execution Bind Point.

Previous approvals, governance activities or earlier runtime evaluations shall not be assumed to remain valid.

Execution authority is determined using the Runtime Context that exists immediately before institutional consequence formation.

This ensures that runtime admissibility reflects the current operational state rather than historical assumptions.

3.8 Architectural Objective

Taken together, these principles establish a runtime architecture whose purpose is narrowly defined:

To provide deterministic, implementation-independent evaluation of whether delegated authority remains legitimately exercisable immediately before AI-generated clinical documentation creates an institutional consequence.

4. High-Level Runtime Architecture

The ASVH Runtime Architecture defines an implementation-independent reference model for evaluating runtime admissibility within AI-enabled ambient scribing workflows.

The architecture deliberately separates governance, AI generation, clinical accountability and Runtime Authority into independent architectural responsibilities. Each component performs a single, clearly defined function and communicates with adjacent components through explicit execution boundaries.

The architecture does not prescribe the internal implementation of ambient scribing products, EPR (EPR) systems or Runtime Authority implementations. Instead, it illustrates the logical point at which deterministic runtime verification occurs immediately before institutional consequence formation.

The objective of the architecture is to demonstrate how an independent Runtime Authority can complement existing healthcare systems without replacing governance, clinical judgement or AI-assisted documentation generation.

Figure 4.1 – High-Level Runtime Architecture

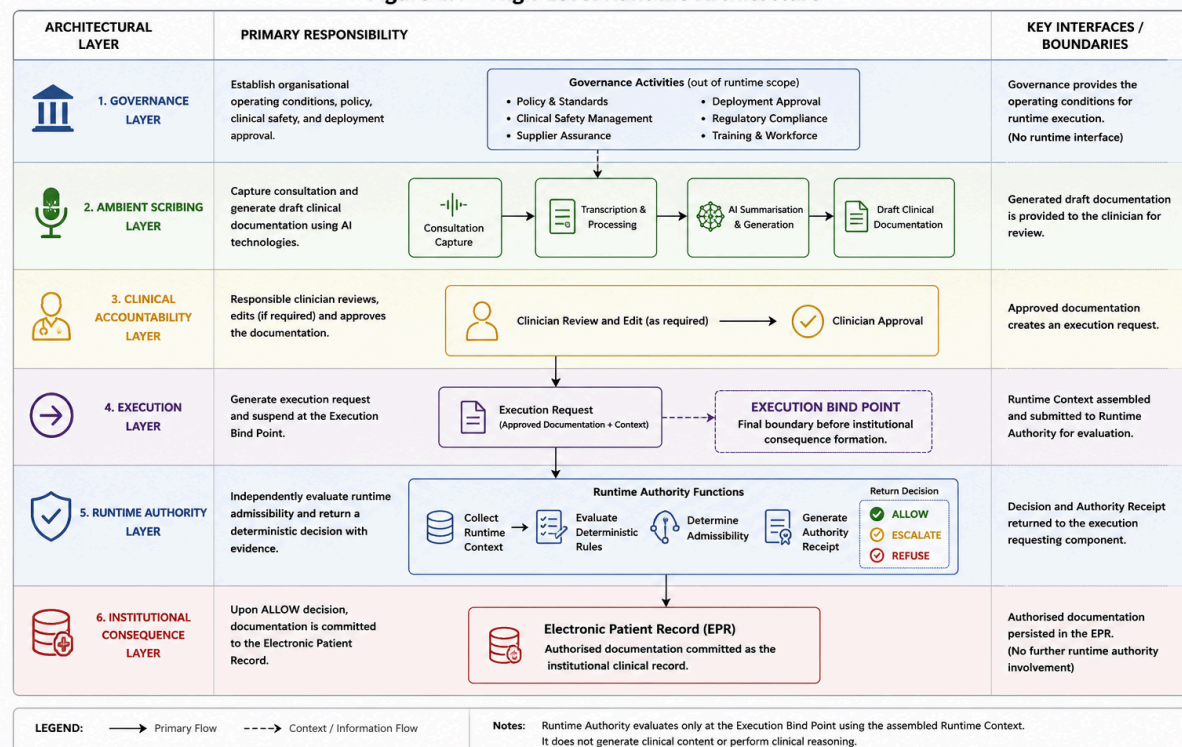


Figure 4.1 – High-Level Runtime Architecture

Figure 4.1 presents the logical architecture of the **NHS Ambient Scribing Runtime Verification Harness (ASVH)** and illustrates the separation of responsibilities between governance, AI-assisted documentation generation, clinical accountability, runtime verification and institutional consequence.

The architecture is organised into six logical layers. Governance establishes the organisational policies, clinical safety processes and deployment approvals under which AI-enabled ambient scribing operates but does not participate in runtime execution. The Ambient Scribing Layer captures the clinical consultation and generates draft documentation, which is subsequently reviewed and approved by the responsible clinician within the Clinical Accountability Layer.

Following clinician approval, an execution request is suspended at the **Execution Bind Point**, representing the final architectural boundary before institutional consequence formation. At this point, the Runtime Context is assembled and submitted to the **Independent Runtime Authority**, which performs deterministic evaluation of runtime admissibility. Based on the current operational state, the Runtime Authority returns one of three decisions—**ALLOW**, **ESCALATE** or **REFUSE**—and generates an **Authority Receipt** containing the evidence supporting the decision.

Only an **ALLOW** decision permits documentation to be committed to the **EPR (EPR)**, where it becomes part of the institutional clinical record. By positioning the Runtime Authority immediately before consequence formation, the architecture preserves the independence of governance, clinician judgement and ambient scribing systems while introducing a deterministic execution-time verification layer that complements existing NHS clinical safety processes.

The architecture consists of six logical layers.

Collectively these layers define the complete execution path from organisational governance through to institutional consequence while preserving independent ownership of each architectural responsibility.

4.1 Governance Layer

The Governance Layer establishes the organisational environment within which AI-enabled ambient scribing operates.

Typical responsibilities include:

- organisational policy;
- clinical safety management;
- supplier assurance;
- deployment approval;
- regulatory compliance;
- workforce training.

These responsibilities exist independently of runtime execution and remain outside the Runtime Authority.

4.2 Ambient Scribing Layer

The Ambient Scribing Layer captures the clinical consultation and generates draft clinical documentation.

Typical responsibilities include:

- consultation capture;
- speech recognition;
- transcription;
- language model execution;
- clinical summarisation;
- document generation.

The Runtime Authority neither participates in nor influences these activities.

4.3 Clinical Accountability Layer

Before documentation can be committed to the EPR, a responsible clinician reviews the generated output, makes any necessary amendments and explicitly approves the documentation.

Clinical accountability remains entirely with the responsible clinician.

The Runtime Authority performs no clinical reasoning, and neither validates nor modifies clinical content.

4.4 Execution Layer

Following clinician approval, an execution request is generated.

Execution is intentionally suspended at the **Execution Bind Point**, representing the final architectural boundary before institutional consequence formation.

At this point, the Runtime Context is assembled and submitted for independent evaluation.

No execution proceeds beyond the Execution Bind Point until the Runtime Authority has returned a deterministic decision.

4.5 Runtime Authority Layer

The Runtime Authority independently evaluates whether delegated authority remains legitimately exercisable immediately before execution.

Its responsibilities include:

- receiving the Runtime Context;
- evaluating deterministic runtime rules;
- determining runtime admissibility;
- generating execution evidence;
- producing an Authority Receipt;
- returning one of three deterministic outcomes:
 - **ALLOW**
 - **ESCALATE**
 - **REFUSE**

The Runtime Authority neither generates clinical documentation nor performs clinical reasoning.

Its sole responsibility is the evaluation of execution admissibility.

4.6 Institutional Consequence Layer

Following an **ALLOW** decision, the approved documentation is committed to the EPR.

The EPR remains the authoritative institutional record.

The Runtime Authority neither stores nor modifies clinical information following execution.

4.7 Architectural Rationale

The placement of the Runtime Authority within the execution sequence is a deliberate architectural decision.

Alternative placements were considered and rejected for the following reasons.

Before AI Generation

Placing Runtime Authority before AI generation would require it to participate in the production of clinical documentation, blurring the separation between content generation and execution assurance.

Before Clinician Review

Placing Runtime Authority before clinician review would require evaluation of documentation that has not yet been accepted by the responsible clinician.

Professional accountability would therefore remain unresolved.

After EPR Commitment

Placing Runtime Authority after documentation has been committed to the EPR would reduce runtime verification to retrospective audit.

At that point the institutional consequence has already occurred and execution can no longer be prevented.

The **Execution Bind Point** therefore represents the only architectural location at which runtime admissibility can be independently evaluated immediately before consequence formation.

This positioning preserves the independence of Runtime Authority while ensuring that evaluation is performed using the complete operational context available at the moment of execution.

4.8 Relationship to Existing Architectures

Existing ambient scribing architectures primarily describe the generation, review and storage of clinical documentation.

Typical architectural components include:

- consultation capture;
- transcription;
- language model processing;
- clinician review;
- EPR integration.

ASVH introduces an additional architectural responsibility.

Rather than extending the ambient scribing platform itself, the architecture introduces an **Independent Runtime Authority** positioned immediately before institutional consequence formation.

This additional architectural layer evaluates whether execution remains admissible under the current runtime conditions.

The architecture therefore complements existing ambient scribing platforms, governance processes and EPR systems without replacing any of them.

4.9 Architectural Separation

The ordering of the architectural layers is intentional.

Each layer owns a single architectural responsibility.

Architectural Layer	Primary Responsibility
Governance	Establish organisational operating conditions
Ambient Scribing	Generate clinical documentation
Clinical Accountability	Review and approve documentation
Execution	Suspend execution at the Execution Bind Point
Runtime Authority	Determine runtime admissibility
EPR	Persist the authorised institutional record

No layer assumes responsibilities belonging to another.

This separation of concerns ensures that Runtime Authority evaluates only execution admissibility while governance, AI generation, clinician accountability and institutional record management remain independent architectural responsibilities.

4.10 Chapter Summary

The High-Level Runtime Architecture establishes a clear separation between AI generation, clinical accountability, governance and runtime execution.

By introducing an independent Runtime Authority at the Execution Bind Point, the architecture demonstrates how deterministic runtime admissibility can be evaluated immediately before institutional consequence formation without altering existing ambient scribing platforms, EPR systems or governance processes.

This architecture provides the foundation upon which the subsequent chapters define architectural components, Runtime Context, rule evaluation, Authority Receipt generation and implementation-independent deployment.

5. Architectural Components

The ASVH Runtime Architecture is composed of a number of independent architectural components that collaborate to support deterministic runtime evaluation immediately before institutional consequence formation.

Each component has a clearly defined responsibility and communicates through explicit interfaces. No component assumes responsibilities allocated to another, ensuring that governance, AI generation, clinical accountability and runtime verification remain architecturally independent.

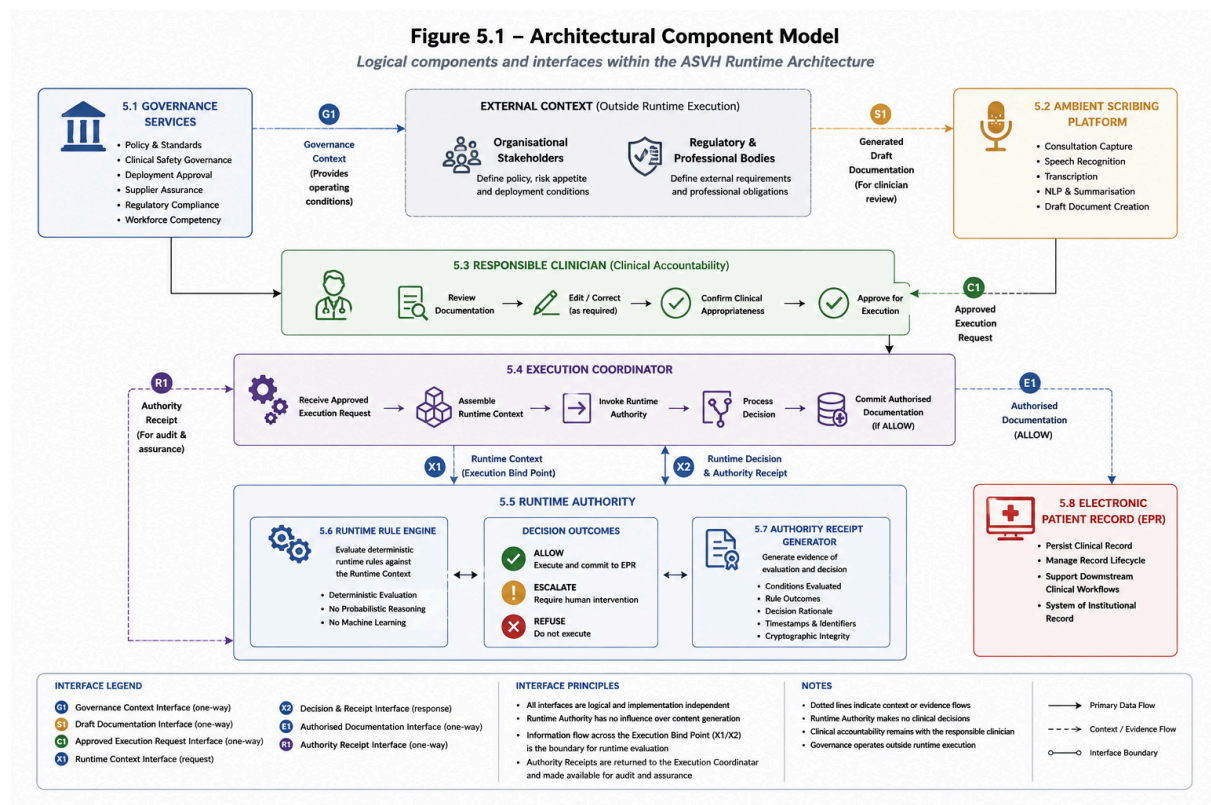


Figure 5.1 – Architectural Component Model

Figure 5.1 illustrates the principal architectural components that comprise the ASVH Runtime Architecture and the logical interfaces through which they interact during execution. Unlike the high-level architectural view presented in Figure 4.1, this figure focuses on the individual responsibilities of each component and the flow of information between them.

The architecture begins with **Governance Services**, which establish the organisational policies, clinical safety processes and deployment conditions under which ambient scribing

operates. These governance activities provide the operational context for the workflow but remain external to runtime execution.

The **Ambient Scribing Platform** captures the clinical consultation and generates draft clinical documentation, which is subsequently reviewed and approved by the **Responsible Clinician**. Professional accountability remains with the clinician, whose approval authorises progression to the execution phase but does not itself determine execution admissibility.

The **Execution Coordinator** receives the approved execution request, assembles the Runtime Context and invokes the **Independent Runtime Authority**. Within the Runtime Authority, the **Runtime Rule Engine** evaluates the Runtime Context against deterministic runtime rules defined within the ASVH specification. Based on this evaluation, the Runtime Authority returns one of three deterministic outcomes—**ALLOW**, **ESCALATE** or **REFUSE**—and the **Authority Receipt Generator** records the evidence supporting the decision.

Where an **ALLOW** decision is returned, the Execution Coordinator commits the authorised documentation to the **EPR (EPR)**, where it becomes part of the institutional clinical record. The generated Authority Receipt remains available as an independent engineering evidence artefact supporting audit, traceability and deterministic replay.

The labelled interfaces shown in the figure demonstrate that communication between architectural components occurs through explicit, implementation-independent boundaries. This separation of responsibilities ensures that governance, AI generation, clinical accountability, runtime evaluation and institutional record management remain independently owned architectural concerns while collaborating to support deterministic execution-time verification.

5.1 Governance Services

Governance Services establish the organisational framework within which AI-enabled ambient scribing operates.

Typical responsibilities include:

- organisational policy;
- clinical safety governance;
- deployment approval;
- supplier assurance;
- workforce competency;
- regulatory compliance.

Governance Services define the operating conditions for deployment but do not participate in runtime execution.

5.2 Ambient Scribing Platform

The Ambient Scribing Platform captures the clinical consultation and generates draft clinical documentation.

Typical responsibilities include:

- consultation capture;
- speech recognition;
- transcription;
- natural language processing;
- clinical summarisation;
- draft document creation.

The internal implementation of the ambient scribing platform is outside the scope of ASVH.

The Runtime Authority treats the platform as an external component and does not inspect or influence its internal processing.

5.3 Responsible Clinician

The Responsible Clinician remains accountable for reviewing and approving AI-generated clinical documentation.

Responsibilities include:

- reviewing generated documentation;
- correcting inaccuracies;
- confirming clinical appropriateness;
- approving documentation for execution.

Clinical approval represents professional accountability but does not, by itself, authorise institutional execution.

Execution authority is evaluated independently by the Runtime Authority.

5.4 Execution Coordinator

The Execution Coordinator manages the transition between clinician approval and institutional execution.

Its responsibilities include:

- receiving the approved execution request;
- assembling the Runtime Context;
- invoking the Runtime Authority;
- processing the returned decision;
- committing authorised documentation following an **ALLOW** decision.

The Execution Coordinator performs no runtime evaluation itself.

5.5 Runtime Authority

The Runtime Authority is the central architectural component within ASVH.

Its sole responsibility is to determine whether delegated authority remains legitimately exercisable immediately before institutional consequence formation.

To achieve this, it:

- receives the Runtime Context;
- evaluates deterministic runtime rules;
- determines runtime admissibility;
- generates an Authority Receipt;
- returns a deterministic execution decision.

The Runtime Authority does not:

- generate clinical documentation;
- perform clinical reasoning;
- modify clinical records;
- establish organisational policy;
- approve clinical content.

Its responsibility is limited to execution-time admissibility.

5.6 Runtime Rule Engine

The Runtime Rule Engine evaluates the Runtime Context against the deterministic runtime rules defined within **ASVH-SPEC-001**.

Each rule evaluates a single runtime condition.

Collectively, these evaluations determine whether execution remains admissible.

The rule engine performs no probabilistic reasoning and does not apply machine learning or adaptive optimisation.

5.7 Authority Receipt Generator

The Authority Receipt Generator records the evidence supporting every runtime decision.

The receipt provides:

- evaluated runtime conditions;
- rule outcomes;
- execution decision;
- supporting evidence;
- timestamps;
- execution identifiers.

Authority Receipts support:

- audit;
- deterministic replay;
- traceability;
- engineering verification.

The Authority Receipt is an engineering evidence artefact and does not form part of the clinical record.

5.8 EPR

The EPR (EPR) represents the institutional destination of authorised clinical documentation.

The EPR is responsible for:

- persistence of clinical records;
- institutional record management;
- downstream clinical workflows.

The Runtime Authority has no responsibility for the internal operation of the EPR.

Its involvement concludes once a deterministic runtime decision has been returned.

5.9 Component Relationships

Each architectural component has a single, clearly defined responsibility.

Component	Primary Responsibility
Governance Services	Define organisational operating conditions
Ambient Scribing Platform	Generate clinical documentation
Responsible Clinician	Review and approve documentation
Execution Coordinator	Coordinate runtime evaluation and execution
Runtime Authority	Determine runtime admissibility
Runtime Rule Engine	Evaluate deterministic runtime rules
Authority Receipt Generator	Produce execution evidence
EPR	Persist authorised clinical documentation

This separation of responsibilities minimises architectural coupling, supports implementation independence and enables Runtime Authority to operate as a discrete execution-time verification component.

5.10 Chapter Summary

The ASVH Runtime Architecture is composed of independent components, each responsible for a single aspect of the execution workflow. By separating governance, AI generation, clinical accountability, execution coordination, runtime evaluation and institutional record management, the architecture preserves clear responsibility boundaries while enabling deterministic runtime verification immediately before institutional consequence formation.

6. Runtime Context

The Runtime Context represents the complete set of operational information evaluated by the Runtime Authority immediately before institutional consequence formation.

Rather than relying upon historical approvals or assumptions, the Runtime Authority evaluates the current operational state that exists at the Execution Bind Point. This ensures that execution admissibility is determined using contemporaneous evidence rather than previous decisions.

The Runtime Context is assembled by the Execution Coordinator immediately prior to runtime evaluation and remains immutable throughout the evaluation process.

The Runtime Authority neither creates nor modifies the Runtime Context. Its responsibility is limited to deterministic evaluation of the supplied evidence.

6.1 Purpose of the Runtime Context

The Runtime Context provides the evidence required to determine whether delegated authority remains legitimately exercisable.

It enables Runtime Authority to evaluate:

- whether the approved documentation remains eligible for execution;
- whether required runtime conditions remain satisfied;
- whether mandatory organisational constraints remain valid;
- whether execution should be allowed, escalated or refused.

The Runtime Context therefore represents the operational state immediately preceding execution rather than the historical process that led to clinician approval.

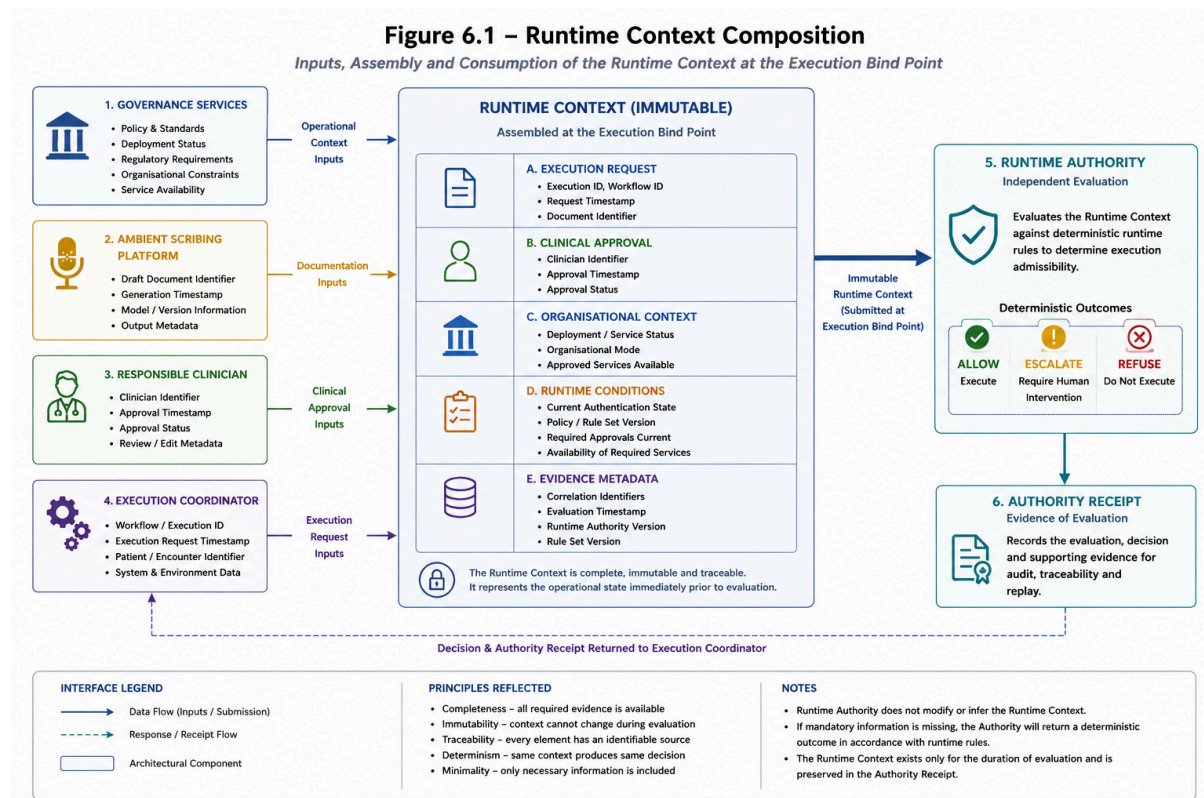


Figure 6.1 – Runtime Context Composition

Figure 6.1 illustrates the composition of the **Runtime Context**, the complete and immutable set of operational information evaluated by the Runtime Authority immediately before institutional consequence formation.

The figure demonstrates that the Runtime Context is assembled at the **Execution Bind Point** from multiple independent architectural sources. Governance Services provide the operational policies and organisational constraints under which execution is permitted. The Ambient Scribing Platform contributes metadata relating to the generated clinical documentation, while the Responsible Clinician provides evidence of clinical review and approval. The Execution Coordinator contributes execution-specific information required to identify and process the proposed transaction.

These independent inputs are consolidated into a single **immutable Runtime Context**, which contains five logical categories of information:

- **Execution Request** – identifies the proposed execution.
- **Clinical Approval** – records clinician authorisation and approval metadata.
- **Organisational Context** – captures the operational state of the organisation at the time of execution.
- **Runtime Conditions** – contains the operational values evaluated by deterministic runtime rules.
- **Evidence Metadata** – records the identifiers, timestamps and version information necessary to support audit, traceability and deterministic replay.

Once assembled, the Runtime Context is submitted to the **Independent Runtime Authority**, which evaluates the supplied evidence against deterministic runtime rules to determine execution admissibility. The Runtime Authority neither creates nor modifies the Runtime Context; it evaluates only the information presented at the Execution Bind Point.

The evaluation results in one of three deterministic outcomes—**ALLOW**, **ESCALATE** or **REFUSE**—together with an **Authority Receipt** that records the evaluated evidence, decision and supporting metadata. The Authority Receipt is returned to the Execution Coordinator and retained as an independent engineering evidence artefact supporting audit, traceability and deterministic replay.

Figure 6.1 reinforces a fundamental architectural principle of the ASVH Runtime Architecture: **Runtime Authority is an evidence consumer rather than an evidence producer**. It performs deterministic evaluation of a complete, immutable Runtime Context without participating in governance, AI generation, clinical judgement or execution coordination, thereby preserving the independence and implementation-agnostic nature of the architecture.

6.2 Runtime Context Principles

The Runtime Context is governed by five principles.

Completeness

All information required to evaluate runtime admissibility shall be available at the point of execution.

Immutability

Once assembled, the Runtime Context shall remain unchanged throughout evaluation.

Traceability

Every Runtime Context element shall be traceable to an identifiable source.

Determinism

An identical Runtime Context shall always produce an identical Runtime Authority decision.

Minimality

Only information required for deterministic evaluation shall be included.

6.3 Runtime Context Categories

For the purposes of ASVH, the Runtime Context consists of several logical categories of information.

Execution Request

Information describing the proposed execution.

Typical examples include:

- execution identifier;
 - workflow identifier;
 - request timestamp;
 - document identifier.
-

Clinical Approval

Evidence that the responsible clinician has approved execution.

Typical examples include:

- clinician identifier;
 - approval timestamp;
 - approval status.
-

Organisational Context

Information describing the operational state of the healthcare organisation.

Typical examples include:

- deployment status;
 - organisational operating mode;
 - approved service availability.
-

Runtime Conditions

The current operational conditions evaluated by deterministic runtime rules.

Examples include:

- clinician remains authenticated;

- patient record remains available;
 - runtime policy version remains valid;
 - mandatory approvals remain current.
-

Evidence Metadata

Supporting metadata required for deterministic replay and audit.

Typical examples include:

- correlation identifiers;
 - evaluation timestamp;
 - Runtime Authority version;
 - rule set version.
-

6.4 Runtime Context Lifecycle

The Runtime Context progresses through four logical stages.

1. Execution request received.
2. Runtime Context assembled.
3. Runtime Authority evaluation.
4. Authority Receipt generated.

The Runtime Context exists only for the duration of runtime evaluation and is preserved within the resulting Authority Receipt to support audit and deterministic replay.

6.5 Architectural Responsibility

The Runtime Context is assembled by the **Execution Coordinator**.

The Runtime Authority:

- does not generate Runtime Context;
- does not modify Runtime Context;
- does not infer missing information.

Instead, it evaluates the Runtime Context exactly as supplied.

If mandatory information is absent, the Runtime Authority shall return a deterministic outcome in accordance with the runtime rules defined within **ASVH-SPEC-001**.

6.6 Relationship to Runtime Rules

The Runtime Context provides the evidence against which deterministic runtime rules are evaluated.

Each runtime rule consumes only the Runtime Context elements necessary to evaluate its assigned condition.

This separation between **Runtime Context** (evidence) and **Runtime Rules** (evaluation logic) ensures that evidence collection remains independent of decision-making and supports implementation-independent architectural design.

6.7 Chapter Summary

The Runtime Context provides the evidential foundation upon which Runtime Authority performs deterministic evaluation.

By assembling a complete, immutable and traceable representation of the operational state immediately before execution, the architecture ensures that runtime admissibility is determined using current evidence rather than historical assumptions. This separation between evidence collection and deterministic rule evaluation is fundamental to the implementation-independent design of the ASVH Runtime Architecture.

7. Runtime Evaluation Sequence

The Runtime Evaluation Sequence defines the logical interactions between architectural components from the point at which an execution request is received until a deterministic execution decision is returned.

The sequence represents the operational behaviour of the ASVH Runtime Architecture immediately before institutional consequence formation.

Although the architecture may be implemented using different technologies, the logical sequence of interactions remains implementation independent.

Figure 7.1 – Runtime Evaluation Sequence

Insert Figure 7.1 here

7.1 Sequence Overview

Runtime evaluation begins only after a responsible clinician has approved the AI-generated clinical documentation for execution.

At this stage, documentation has completed clinical review but has not yet been committed to the EPR.

Execution is suspended at the **Execution Bind Point**, allowing Runtime Authority to determine whether delegated authority remains legitimately exercisable before institutional consequence occurs.

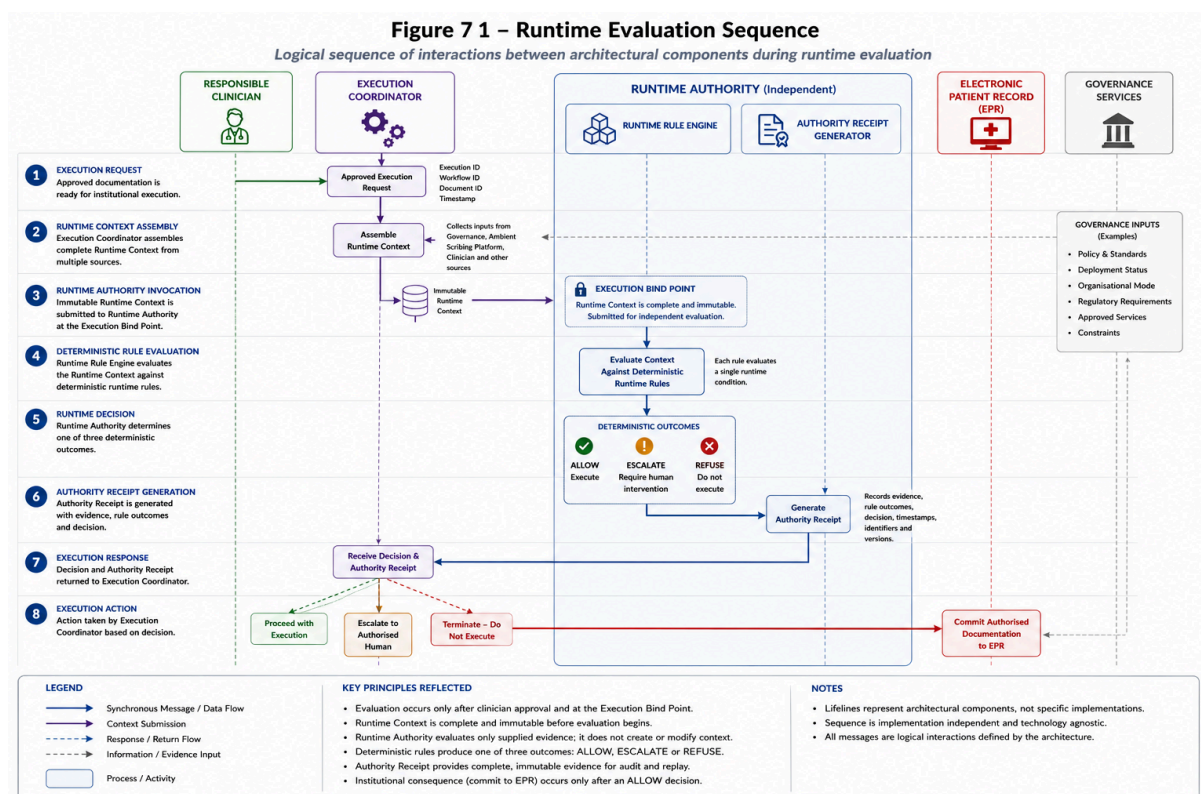


Figure 7.1 – Runtime Evaluation Sequence

Figure 7.1 illustrates the logical sequence of interactions that occur during runtime evaluation within the ASVH Runtime Architecture. It demonstrates how independently owned architectural components collaborate to determine execution admissibility immediately before AI-generated clinical documentation is committed to the EPR (EPR).

The sequence begins once a **Responsible Clinician** has reviewed and approved the generated clinical documentation. Clinical approval authorises progression to runtime evaluation but does not itself permit institutional execution. Instead, the Execution Coordinator receives an approved **execution request** and assembles the complete **Runtime Context** from the relevant architectural sources.

Once the Runtime Context has been assembled and verified as complete, it is submitted to the **Independent Runtime Authority** at the **Execution Bind Point**. The Runtime Authority performs deterministic evaluation by applying the Runtime Context to the predefined runtime rules. Each rule evaluates a single runtime condition, and together these evaluations determine whether execution remains admissible under the current operational state.

The Runtime Authority returns one of three deterministic outcomes:

- **ALLOW** – execution remains admissible and may proceed;
- **ESCALATE** – authorised human intervention is required before execution;
- **REFUSE** – execution is not currently admissible and must not proceed.

Following determination of the runtime outcome, the Runtime Authority generates an **Authority Receipt** containing the evaluated Runtime Context, rule outcomes, execution decision and supporting evidence. The decision and Authority Receipt are returned to the **Execution Coordinator**, which performs the appropriate execution action based on the returned decision. Only an **ALLOW** decision permits documentation to be committed to the **EPR**, thereby creating the institutional clinical record.

The figure also highlights several important architectural characteristics. Runtime evaluation occurs only after clinician approval and immediately before institutional consequence formation. The Runtime Context remains complete and immutable throughout evaluation, while the Runtime Authority functions solely as an independent evaluator of supplied evidence. Governance Services provide the organisational operating conditions for execution but remain external to the runtime sequence itself.

Figure 7.1 demonstrates that runtime admissibility is achieved through a deterministic sequence of independent architectural interactions rather than through probabilistic reasoning or retrospective audit. By clearly separating evidence assembly, runtime evaluation, execution coordination and institutional record management, the sequence preserves the architectural principles of independence, determinism and evidential accountability established throughout the ASVH Runtime Architecture.

7.2 Step 1 – Execution Request

The Execution Coordinator receives an approved execution request.

The request identifies the proposed institutional action and provides the information necessary to begin runtime evaluation.

Typical information includes:

- execution identifier;
- workflow identifier;
- document identifier;
- request timestamp.

No runtime decision has yet been made.

7.3 Step 2 – Runtime Context Assembly

The Execution Coordinator assembles the Runtime Context.

Information is collected from the architectural components defined in Chapter 5 and organised according to the Runtime Context model described in Chapter 6.

The completed Runtime Context represents the operational state immediately preceding execution.

Once assembled, the Runtime Context becomes immutable for the duration of evaluation.

7.4 Step 3 – Runtime Authority Invocation

The immutable Runtime Context is submitted to the Independent Runtime Authority.

The Runtime Authority accepts the supplied evidence exactly as presented.

It neither modifies the Runtime Context nor attempts to infer missing information.

If mandatory information is unavailable, the Runtime Authority returns the deterministic outcome defined by the applicable runtime rules.

7.5 Step 4 – Deterministic Rule Evaluation

The Runtime Rule Engine evaluates the Runtime Context against the deterministic runtime rules defined within **ASVH-SPEC-001**.

Each rule evaluates a single runtime condition.

Collectively, the evaluated rules determine whether execution remains admissible.

No probabilistic reasoning, machine learning or adaptive optimisation occurs during this process.

7.6 Step 5 – Runtime Decision

Following completion of rule evaluation, the Runtime Authority determines one of three deterministic execution outcomes.

ALLOW

Execution remains admissible.

The approved documentation may proceed to institutional execution.

ESCALATE

Runtime evaluation has identified conditions requiring authorised human intervention before execution may proceed.

Execution remains suspended pending resolution.

REFUSE

Execution is not currently admissible.

Institutional consequence formation is prevented.

7.7 Step 6 – Authority Receipt Generation

Following determination of the execution outcome, the Runtime Authority generates an Authority Receipt.

The receipt records:

- evaluated Runtime Context;
- rule outcomes;
- execution decision;
- supporting evidence;
- timestamps;
- Runtime Authority version;
- rule set version.

The Authority Receipt provides the evidence required to support audit, traceability and deterministic replay.

7.8 Step 7 – Execution Response

The Runtime Authority returns the execution decision together with the Authority Receipt to the Execution Coordinator.

The Execution Coordinator performs no additional runtime evaluation.

Its responsibility is limited to acting upon the returned decision.

Where the decision is **ALLOW**, execution proceeds.

Where the decision is **ESCALATE** or **REFUSE**, execution remains suspended or is terminated in accordance with organisational policy.

7.9 Sequence Characteristics

The Runtime Evaluation Sequence exhibits the following architectural characteristics:

- deterministic execution;
- immutable Runtime Context;
- implementation independence;
- explicit execution boundaries;
- complete evidential traceability;
- independent Runtime Authority;
- reproducible execution decisions.

These characteristics ensure that identical Runtime Contexts always produce identical runtime outcomes.

7.10 Architectural Significance

The Runtime Evaluation Sequence defines the operational behaviour of the Runtime Authority independently of any particular software implementation.

By separating Runtime Context assembly, deterministic rule evaluation and execution coordination into independent architectural responsibilities, the sequence preserves the architectural principles established in Chapters 3 and 4 while ensuring that execution admissibility is determined immediately before institutional consequence formation.

7.11 Architectural Rationale

The Runtime Evaluation Sequence is intentionally ordered to preserve the independence of governance, clinical accountability and Runtime Authority while ensuring that execution

admissibility is determined using the complete operational context available immediately before institutional consequence formation.

Alternative execution sequences were considered during the development of the ASVH architecture.

For example:

- evaluating Runtime Authority before clinician approval would require runtime evaluation of documentation that has not yet been clinically accepted;
- evaluating Runtime Authority after commitment to the EPR would reduce runtime verification to retrospective audit;
- allowing Runtime Authority to assemble or modify the Runtime Context would compromise the separation between evidence collection and evidence evaluation.

The adopted sequence therefore preserves four architectural principles:

- clinician accountability remains independent;
- Runtime Context is complete before evaluation begins;
- Runtime Authority evaluates rather than constructs evidence;
- institutional consequence cannot occur until deterministic evaluation has completed.

These design decisions ensure that runtime admissibility is evaluated using the operational state that exists immediately before execution while preserving clear separation of architectural responsibilities.

7.12 Chapter Summary

The Runtime Evaluation Sequence defines the implementation-independent interactions through which the ASVH Runtime Architecture determines execution admissibility immediately before institutional consequence formation.

Beginning with an approved execution request and concluding with the return of a deterministic execution decision and Authority Receipt, the sequence demonstrates how independent architectural components collaborate while preserving clear separation of responsibilities between governance, clinical accountability, execution coordination and Runtime Authority.

The sequence has been deliberately structured to ensure that runtime evaluation occurs only after clinician approval, using a complete and immutable Runtime Context assembled at the Execution Bind Point. By positioning Runtime Authority immediately before commitment to the EPR, the architecture enables execution-time verification without influencing clinical judgement, AI-generated documentation or organisational governance.

Together, the Runtime Evaluation Sequence and its supporting architectural rationale demonstrate why deterministic runtime evaluation is positioned as the final verification activity before institutional consequence formation, preserving the core architectural principles of independence, determinism, implementation independence and evidential accountability.

7.13 Worked Runtime Evaluation Examples

The following examples illustrate how the Runtime Evaluation Sequence defined in this chapter produces deterministic execution outcomes using different Runtime Contexts. These examples are illustrative only and do not define additional Runtime Rules beyond those specified in ASVH-SPEC-001.

Each example follows the same architectural sequence:

1. Clinician approval.
2. Execution Request received.
3. Runtime Context assembled.
4. Runtime Authority evaluation.
5. Runtime Decision returned.
6. Authority Receipt generated.
7. Execution action performed.

The examples demonstrate that different Runtime Contexts produce different deterministic outcomes while preserving identical architectural behaviour.

Example A – ALLOW

Scenario

A clinician has approved AI-generated documentation. All required runtime conditions remain satisfied immediately before commitment to the Electronic Patient Record.

Runtime Condition	Result
Clinician authenticated	PASS
Patient record available	PASS

Runtime policy valid	PASS
----------------------	------

Required approvals present	PASS
----------------------------	------

Clinical session active	PASS
-------------------------	------

Runtime Decision

ALLOW

The Runtime Authority determines that delegated authority remains legitimately exercisable. The approved documentation is committed to the Electronic Patient Record.

The Authority Receipt records the evaluated Runtime Context, Runtime Rule outcomes, Runtime Decision and supporting metadata.

Example B – ESCALATE

Scenario

The clinician has approved the documentation. During runtime evaluation, the Runtime Authority determines that a mandatory escalation contact is unavailable.

Runtime Condition	Result
-------------------	--------

Clinician authenticated	PASS
-------------------------	------

Patient record available	PASS
--------------------------	------

Runtime policy valid	PASS
----------------------	------

Escalation contact available	FAIL
------------------------------	------

Clinical session active	PASS
-------------------------	------

Runtime Decision

ESCALATE

Execution is suspended pending authorised human intervention.

No documentation is committed to the Electronic Patient Record until a subsequent Runtime Evaluation produces an ALLOW decision.

A new Runtime Context must be assembled before re-evaluation.

Example C – REFUSE

Scenario

The clinician has approved the documentation. During runtime evaluation the patient record is no longer available due to system state.

Runtime Condition	Result
Clinician authenticated	PASS
Patient record available	FAIL
Runtime policy valid	PASS
Required approvals present	PASS
Clinical session active	PASS

Runtime Decision

REFUSE

The Runtime Authority determines that execution is no longer admissible.

Institutional consequence formation is prevented.

The Authority Receipt records the evaluated Runtime Context and evidence supporting the refusal.

Architectural Observation

These examples demonstrate that the Runtime Authority does not alter clinical documentation, clinician judgement or organisational governance. It evaluates only the Runtime Context presented at the Execution Bind Point and deterministically returns one of three possible execution outcomes: ALLOW, ESCALATE or REFUSE.

The architectural sequence remains identical in every case. Only the Runtime Context changes, and therefore only the Runtime Decision changes.

8. Runtime Decision Architecture

The Runtime Decision Architecture defines the logical process through which the Runtime Authority transforms an immutable Runtime Context into a single deterministic execution decision.

Unlike the Runtime Rule Specifications defined within **ASVH-SPEC-001**, this chapter describes the architectural relationship between Runtime Context, Runtime Rule evaluation and Runtime Authority decision formation. It does not prescribe the implementation of individual rules.

The Runtime Decision Architecture ensures that every execution request produces one—and only one—deterministic Runtime Decision based solely upon the Runtime Context presented at the Execution Bind Point.

8.1 Architectural Objective

The objective of the Runtime Decision Architecture is to ensure that execution admissibility is determined consistently, transparently and independently of implementation technology.

The architecture achieves this by separating:

- Runtime Context assembly;
- Runtime Rule evaluation;
- Runtime Decision formation; and
- Authority Receipt generation.

Each activity performs a single architectural responsibility and contributes to the overall determinism of the Runtime Authority.

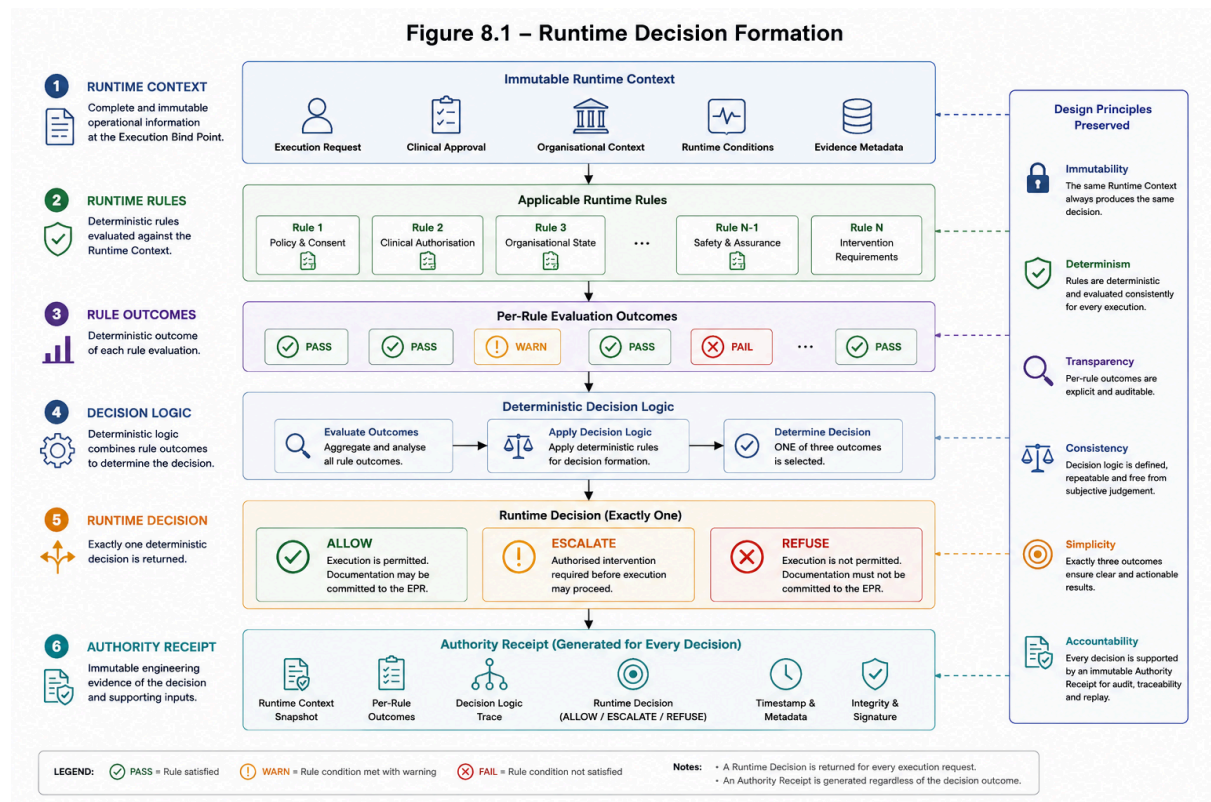


Figure 8.1 – Runtime Decision Formation

Description

Figure 8.1 illustrates the logical process through which the **Independent Runtime Authority** transforms an immutable **Runtime Context** into a single deterministic **Runtime Decision** and the associated **Authority Receipt**.

The process begins with the receipt of a complete and immutable **Runtime Context** assembled by the **Execution Coordinator** at the **Execution Bind Point**. The Runtime Context contains all operational information required to evaluate execution admissibility and remains unchanged throughout the evaluation process.

The Runtime Authority applies the relevant **Runtime Rules** to the supplied Runtime Context. Each rule independently evaluates a specific runtime condition, producing an individual rule outcome that reflects whether the corresponding execution requirement has been satisfied.

The resulting **Rule Outcomes** are then combined by the **Decision Logic** component. Rather than relying on probabilistic reasoning or confidence scoring, the Decision Logic applies deterministic evaluation criteria to produce a single execution decision. This architectural separation between rule evaluation and decision formation preserves transparency, repeatability and implementation independence.

The Runtime Authority returns exactly one of three possible **Runtime Decisions**:

- **ALLOW** – all required runtime conditions remain satisfied and execution may proceed.
- **ESCALATE** – execution is suspended pending authorised human intervention and subsequent re-evaluation.
- **REFUSE** – execution is not currently admissible and institutional consequence formation is prevented.

Regardless of the decision reached, the Runtime Authority generates an **Authority Receipt** containing the Runtime Context snapshot, Runtime Rule outcomes, Decision Logic trace, final Runtime Decision and supporting integrity metadata. This receipt provides the complete engineering evidence necessary for audit, deterministic replay and architectural verification.

The figure also highlights the core design principles preserved throughout the decision process. The Runtime Context remains immutable during evaluation, Runtime Rules are deterministic and independently evaluated, Decision Logic is transparent and repeatable, and every execution request produces both a single Runtime Decision and a corresponding Authority Receipt. These principles ensure that identical Runtime Contexts consistently produce identical execution outcomes while maintaining complete evidential accountability.

Figure 8.1 demonstrates that **Runtime Decision formation is a deterministic engineering process rather than a probabilistic AI decision**. By separating Runtime Context evaluation, rule execution, decision formation and evidence generation into distinct architectural responsibilities, the ASVH Runtime Architecture provides a transparent, implementation-independent model for determining execution admissibility immediately before institutional consequence formation.

8.2 Decision Inputs

Runtime Decision formation consumes only the immutable Runtime Context assembled by the Execution Coordinator.

The Runtime Authority does not:

- request additional information;
- infer missing evidence;
- modify Runtime Context values;
- access external systems during evaluation.

Every decision is therefore derived exclusively from the Runtime Context presented at the Execution Bind Point.

8.3 Decision Formation

The Runtime Rule Engine evaluates each applicable Runtime Rule using the supplied Runtime Context.

Each rule produces an individual engineering outcome.

The Runtime Authority then combines these rule outcomes to determine a single Runtime Decision.

At no point does the Runtime Authority apply probabilistic reasoning, confidence scoring or adaptive optimisation.

Decision formation remains entirely deterministic.

8.4 Runtime Decision Outcomes

The Runtime Authority recognises three architectural decision outcomes.

ALLOW

All mandatory runtime conditions remain satisfied.

The requested execution remains admissible and may proceed to institutional consequence formation.

ESCALATE

Runtime evaluation identifies conditions requiring authorised human intervention.

Execution remains suspended until the identified condition has been resolved and the Runtime Authority performs a new evaluation using the current Runtime Context.

REFUSE

Runtime evaluation determines that execution is not currently admissible.

Institutional consequence formation is prevented.

No execution may proceed until a subsequent evaluation produces an alternative Runtime Decision.

8.5 Deterministic Behaviour

Runtime Decision formation satisfies the following deterministic principles.

- identical Runtime Context produces identical Runtime Decisions;
- Runtime Rule evaluation order is consistent;
- decision formation is repeatable;
- Runtime Decisions are reproducible;
- implementation technology does not alter architectural behaviour.

These principles ensure that Runtime Authority behaviour remains objectively verifiable.

8.6 Decision Independence

The Runtime Decision represents an independent architectural determination.

It is independent of:

- AI-generated content;
- clinician judgement;
- governance processes;
- implementation technology;
- deployment architecture.

The Runtime Decision concerns only one architectural question:

Does delegated authority remain legitimately exercisable immediately before institutional consequence formation?

8.7 Architectural Rationale

The Runtime Decision Architecture deliberately separates rule evaluation from decision formation.

This separation enables Runtime Rules to evaluate individual runtime conditions while preserving a single architectural responsibility for Runtime Authority: determining execution admissibility.

Alternative approaches—such as embedding decision logic within individual runtime rules or distributing decision formation across multiple components—were rejected because they would reduce transparency, complicate traceability and weaken deterministic replay.

By centralising Runtime Decision formation within the Runtime Authority, the architecture preserves a clear and explainable relationship between Runtime Context, Runtime Rule outcomes, Authority Receipt evidence and the resulting execution decision.

8.8 Chapter Summary

The Runtime Decision Architecture defines how immutable Runtime Context is transformed into a single deterministic Runtime Decision.

By separating Runtime Context, Runtime Rule evaluation, decision formation and Authority Receipt generation into distinct architectural responsibilities, the ASVH Runtime Architecture preserves implementation independence, deterministic behaviour and complete evidential accountability.

The resulting Runtime Decision provides the authoritative execution outcome upon which subsequent institutional execution is based, while maintaining the separation of responsibilities established throughout the ASVH Runtime Architecture.

9. Authority Receipt Architecture

The Authority Receipt Architecture defines how evidence of runtime evaluation is generated, structured and preserved following deterministic Runtime Authority evaluation.

The Authority Receipt represents the authoritative engineering record of the runtime determination. It provides a complete and immutable record of the evidence evaluated, the deterministic decision reached and the architectural context in which that decision was made.

The Authority Receipt is generated for every runtime evaluation, irrespective of whether the resulting Runtime Decision is **ALLOW**, **ESCALATE** or **REFUSE**.

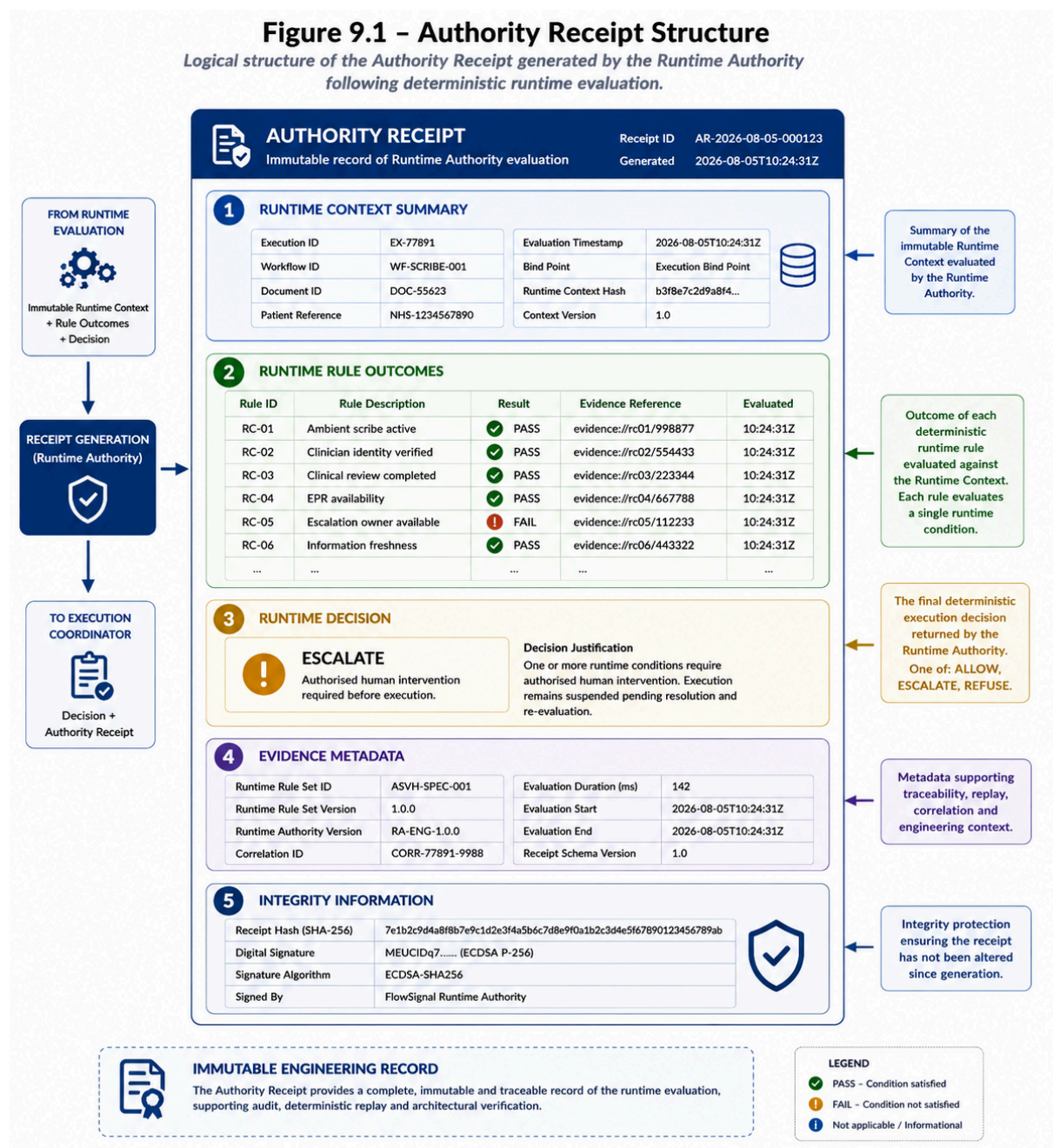


Figure 9.1 – Authority Receipt Structure

Description

Figure 9.1 illustrates the logical structure of the **Authority Receipt**, the engineering evidence artefact generated by the Independent Runtime Authority following deterministic runtime evaluation.

The Authority Receipt provides a complete, immutable and traceable record of how the Runtime Authority reached its execution decision. It is generated immediately after runtime evaluation has completed and is produced irrespective of whether the resulting Runtime Decision is **ALLOW**, **ESCALATE** or **REFUSE**.

The figure shows that receipt generation consumes three architectural inputs: the immutable **Runtime Context**, the evaluated **Runtime Rule Outcomes**, and the final **Runtime Decision**. These inputs are combined to produce a structured Authority Receipt consisting of five logical sections.

The **Runtime Context Summary** records the execution identifiers, workflow information, evaluation timestamp and context metadata describing the operational state evaluated by the Runtime Authority.

The **Runtime Rule Outcomes** section records the result of each deterministic runtime rule, together with supporting evidence references and evaluation timestamps. Each rule evaluates a single runtime condition, allowing the complete decision path to be reconstructed during audit or deterministic replay.

The **Runtime Decision** section records the single authoritative outcome returned by the Runtime Authority—**ALLOW**, **ESCALATE** or **REFUSE**—together with the engineering justification supporting that decision.

The **Evidence Metadata** section captures supporting information required for traceability and engineering verification, including Runtime Rule Set identifiers, Runtime Authority version information, correlation identifiers and evaluation timing.

The **Integrity Information** section protects the Authority Receipt against unauthorised modification by recording integrity-related information such as receipt hashes, digital signatures and associated verification metadata. The architecture defines these as logical capabilities and does not prescribe a specific implementation technology.

The completed Authority Receipt is returned to the **Execution Coordinator** together with the Runtime Decision. While the decision determines whether execution may proceed, the Authority Receipt provides the independent engineering evidence supporting that determination. It remains separate from the EPR and is intended to support audit, deterministic replay, architectural verification and evidential accountability rather than clinical record management.

9.1 Architectural Objective

The objective of the Authority Receipt is to provide an implementation-independent evidence model supporting:

- audit;
- deterministic replay;
- architectural traceability;
- engineering verification;
- operational assurance.

The Authority Receipt provides evidence of the Runtime Authority's determination and does not replace the EPR or clinical documentation.

9.2 Authority Receipt Principles

The Authority Receipt is governed by the following architectural principles.

Completeness

The receipt shall contain all information necessary to explain the Runtime Decision.

Immutability

Once generated, the receipt shall not be modified.

Traceability

Every evidence element shall be traceable to its originating Runtime Context.

Determinism

An identical Runtime Context shall produce an equivalent Authority Receipt.

Independence

The receipt records Runtime Authority evaluation only and does not include AI reasoning or clinical judgement.

9.3 Authority Receipt Structure

An Authority Receipt consists of five logical sections.

Runtime Context Summary

A summary of the Runtime Context evaluated during execution.

Typical information includes:

- execution identifier;
 - workflow identifier;
 - evaluation timestamp;
 - Runtime Authority version.
-

Runtime Rule Outcomes

The outcome of each deterministic runtime rule evaluated during execution.

Typical information includes:

- rule identifier;
 - evaluation result;
 - supporting evidence reference;
 - evaluation timestamp.
-

Runtime Decision

The final Runtime Decision returned by the Runtime Authority.

Possible values are:

- **ALLOW**
- **ESCALATE**
- **REFUSE**

The receipt records only the final deterministic decision.

Evidence Metadata

Supporting information required for replay and audit.

Typical information includes:

- correlation identifiers;
- Runtime Rule version;
- Runtime Authority version;
- receipt identifier.

Integrity Information

Information supporting the integrity of the Authority Receipt.

Examples include:

- receipt hash;
- digital signature (where implemented);
- integrity verification metadata.

The architecture defines these as logical capabilities and does not prescribe a particular implementation technology.

9.4 Receipt Generation

Authority Receipt generation occurs immediately after Runtime Decision formation.

The receipt is produced using:

- the immutable Runtime Context;
- Runtime Rule evaluation outcomes;
- the deterministic Runtime Decision;
- evaluation metadata.

Generation of the receipt does not alter the Runtime Decision.

Instead, it provides the evidential record explaining how the Runtime Authority reached its determination.

9.5 Architectural Responsibilities

The Runtime Authority is responsible for generating the Authority Receipt.

The Execution Coordinator is responsible for receiving the completed receipt.

The EPR is not responsible for storing the Authority Receipt unless required by local implementation policy.

The receipt remains an independent engineering artefact rather than a component of the institutional clinical record.

9.6 Relationship to the Runtime Decision

The Authority Receipt documents the Runtime Decision but does not influence it.

The Runtime Decision is completed before receipt generation begins.

This separation ensures that evidence recording remains independent of decision formation and preserves deterministic behaviour.

9.7 Architectural Rationale

The Authority Receipt is intentionally generated as a separate architectural artefact rather than embedding execution evidence within the Runtime Decision itself.

This separation provides several advantages.

First, it distinguishes the operational outcome (**ALLOW**, **ESCALATE** or **REFUSE**) from the evidence supporting that outcome.

Second, it enables deterministic replay without altering the execution workflow.

Third, it allows evidence models to evolve independently of Runtime Decision semantics.

Alternative approaches, such as recording only the final decision or embedding evidence directly within execution messages, were considered but rejected because they reduce explainability, complicate independent audit and weaken architectural traceability.

The adopted architecture therefore treats the Authority Receipt as the definitive engineering record of runtime evaluation.

9.8 Chapter Summary

The Authority Receipt Architecture provides the evidential foundation supporting deterministic Runtime Authority evaluation.

By separating evidence generation from Runtime Decision formation, the architecture preserves implementation independence while ensuring that every execution decision is supported by a complete, immutable and traceable engineering record.

The Authority Receipt enables audit, deterministic replay and architectural verification without altering the execution workflow or assuming responsibility for institutional clinical records.

10. Implementation Independence and Deployment Architecture

The ASVH Runtime Architecture defines an implementation-independent reference architecture for deterministic runtime verification within AI-enabled ambient scribing workflows.

This chapter describes the architectural deployment principles that enable Runtime Authority to integrate with existing healthcare systems while remaining independent of any specific ambient scribing platform, EPR (EPR), programming language or deployment technology.

The objective is not to prescribe implementation, but to define the architectural characteristics that every compliant implementation should preserve.

10.1 Architectural Independence

The Runtime Authority is defined as an independent architectural component.

Its responsibilities remain unchanged regardless of:

- ambient scribing supplier;
- EPR;
- deployment model;
- cloud platform;
- programming language;
- workflow technology.

Implementation choices shall not alter the architectural responsibilities defined within this document.

10.2 Logical Deployment Model

The architecture defines logical responsibilities rather than physical infrastructure.

A Runtime Authority implementation may be deployed as:

- an independent service;
- an API;
- an execution gateway;
- a containerised component;
- an on-premise service;
- a cloud-native service.

Each deployment model shall preserve the architectural separation between:

- Runtime Context assembly;
 - Runtime Authority evaluation;
 - Runtime Decision formation;
 - Authority Receipt generation;
 - institutional execution.
-

10.3 Integration Principles

The Runtime Authority integrates through clearly defined interfaces.

Integration shall not require modification of:

- ambient scribing products;
- clinician workflows;
- EPR systems;
- organisational governance processes.

Instead, existing execution requests are redirected through the Execution Bind Point immediately before institutional consequence formation.

10.4 Execution Bind Point

The Execution Bind Point represents the architectural interface between execution coordination and institutional consequence.

Every implementation shall preserve this boundary.

No institutional consequence shall occur until the Runtime Authority has returned a deterministic Runtime Decision.

This principle remains independent of deployment technology.

10.5 Vendor Independence

The ASVH Runtime Architecture intentionally avoids dependence upon specific technology vendors.

The architecture is compatible with any ambient scribing platform capable of:

- generating clinical documentation;

- supporting clinician approval;
- invoking Runtime Authority before execution.

Similarly, the architecture does not depend upon a specific EPR implementation.

Vendor independence enables Runtime Authority to operate consistently across heterogeneous healthcare environments.

10.6 Scalability Considerations

The reference architecture places no architectural constraints on deployment scale.

Implementations may support:

- individual clinical departments;
- hospital-wide deployment;
- multi-site NHS Trusts;
- Integrated Care Systems;
- national deployment.

Scaling strategies remain implementation concerns and are outside the scope of this architectural specification.

10.7 Architectural Constraints

Although implementation technology is unrestricted, compliant implementations shall preserve the following architectural constraints:

- Runtime Authority remains independent;
- Runtime Context remains immutable during evaluation;
- Runtime Decisions remain deterministic;
- Authority Receipts remain complete and traceable;
- Runtime evaluation occurs only at the Execution Bind Point.

These constraints define architectural compliance independently of implementation technology.

10.8 Architectural Rationale

The ASVH Runtime Architecture deliberately separates architectural definition from implementation technology.

This enables organisations to adopt Runtime Authority without replacing existing ambient scribing platforms or EPR systems.

Alternative approaches—such as embedding runtime evaluation directly within AI applications or integrating decision logic into individual EPR implementations—were considered but rejected because they increase vendor dependency, reduce portability and weaken architectural independence.

By defining Runtime Authority as an implementation-independent architectural component, the ASVH architecture supports interoperability while preserving deterministic execution-time verification across diverse healthcare environments.

10.9 Chapter Summary

The ASVH Runtime Architecture defines a deployment-independent reference model for Runtime Authority integration.

By separating architectural responsibilities from implementation technology, the architecture enables deterministic runtime verification to be deployed alongside existing ambient scribing platforms and EPR systems without altering their internal operation.

This implementation-independent approach ensures that organisations remain free to select deployment technologies appropriate to their operational environment while preserving the architectural principles of independence, determinism, evidential accountability and execution-time verification established throughout this document.

10.10 Implementation Impact

A principal objective of the ASVH Runtime Architecture is to introduce deterministic runtime verification while minimising disruption to existing healthcare systems.

Rather than replacing existing components, the architecture introduces a single additional architectural capability positioned immediately before institutional consequence formation.

The expected implementation impact is summarised below.

Existing Architectural Component	Implementation Impact
NHS Governance Processes	No architectural change required. Governance continues to define deployment policies and clinical safety responsibilities.
AI-enabled Ambient Scribing Platform	No modification to AI generation or documentation workflow required. Existing execution requests are redirected through the Execution Bind Point.

Responsible Clinician Workflow	No change to clinical review or approval responsibilities. Clinical accountability remains unchanged.
Electronic Patient Record (EPR)	No modification to record management or storage architecture required. Documentation is committed only following an ALLOW decision.
Runtime Authority	New implementation-independent architectural component responsible for deterministic runtime evaluation.
Execution Coordinator	Extended to assemble the Runtime Context, invoke the Runtime Authority and process the returned Runtime Decision and Authority Receipt.
Authority Receipt Repository (optional)	Stores engineering evidence separately from the Electronic Patient Record where required by local implementation policy.

Architectural Impact Summary

The introduction of Runtime Authority requires only one additional architectural interaction within the existing execution workflow.

No changes are required to:

- governance processes;
- clinician responsibilities;
- AI-assisted documentation generation;
- Electronic Patient Record design; or
- organisational operating procedures.

Instead, the architecture inserts deterministic runtime verification immediately before institutional consequence formation while preserving the existing separation of responsibilities.

This deployment approach enables Runtime Authority to be adopted incrementally alongside existing ambient scribing platforms and healthcare environments without requiring replacement of established systems or clinical workflows.

11. Conclusion

The ASVH Runtime Architecture defines an implementation-independent reference architecture for deterministic runtime verification within AI-enabled ambient scribing workflows.

Rather than replacing existing governance frameworks, clinical safety processes or ambient scribing platforms, the architecture introduces an additional architectural responsibility: the independent evaluation of runtime admissibility immediately before institutional consequence formation.

Throughout this document, the Runtime Authority has been presented as a distinct architectural component whose sole responsibility is to determine whether delegated authority remains legitimately exercisable at the Execution Bind Point.

This separation preserves the responsibilities of governance, ambient scribing platforms, responsible clinicians and EPR systems while providing a deterministic mechanism for evaluating execution-time admissibility using the current operational context.

The architecture deliberately separates:

- governance from runtime execution;
- AI generation from execution authority;
- clinical accountability from runtime evaluation;
- Runtime Context assembly from Runtime Rule evaluation;
- Runtime Decision formation from Authority Receipt generation;
- architectural definition from implementation technology.

These separations enable the Runtime Authority to remain implementation independent while supporting deterministic execution decisions that are reproducible, explainable and supported by complete engineering evidence.

The Authority Receipt provides an immutable evidential record of every runtime determination, supporting audit, deterministic replay and architectural verification without becoming part of the institutional clinical record.

By preserving explicit architectural boundaries and limiting each component to a single responsibility, the ASVH Runtime Architecture demonstrates how runtime verification can be integrated alongside existing ambient scribing platforms and healthcare systems without requiring changes to clinical workflows or governance processes.

The architecture therefore provides a reference model through which healthcare organisations and technology suppliers can explore deterministic runtime verification as an additional layer of execution assurance immediately before AI-generated clinical documentation creates an institutional consequence.

11.1 Architectural Contribution

The principal contribution of the ASVH Runtime Architecture is the introduction of an **Independent Runtime Authority** positioned at the **Execution Bind Point**.

This architectural responsibility is distinct from:

- governance;
- clinical safety;
- AI-assisted documentation generation;
- clinician approval;
- EPR management.

Instead, it addresses a narrower engineering question:

Does delegated authority remain legitimately exercisable immediately before institutional consequence formation?

By defining this responsibility independently of implementation technology, the architecture establishes a reusable reference model capable of supporting multiple ambient scribing platforms and healthcare environments while preserving deterministic behaviour and evidential accountability.

11.2 Future Development

The ASVH Runtime Architecture provides the architectural foundation for future implementation and interoperability activities.

Future work may include:

- additional runtime rule sets derived from evolving NHS guidance;
- interoperability profiles supporting multiple ambient scribing suppliers;
- integration with diverse EPR systems;
- standardised Runtime Context models;
- implementation profiles aligned with emerging industry standards.

These developments may extend the architecture while preserving the architectural principles defined within this document.

11.3 Final Statement

The ASVH Runtime Architecture demonstrates that deterministic runtime verification can be introduced as an independent architectural capability within AI-enabled ambient scribing workflows.

By evaluating runtime admissibility immediately before institutional consequence formation, the architecture complements existing governance, clinical safety and professional accountability without replacing them.

The result is an implementation-independent architectural model that preserves clear separation of responsibilities while providing deterministic execution-time assurance supported by complete engineering evidence.

Appendix A – Glossary of Terms

A.1 Purpose

This appendix defines the principal architectural terms used throughout the ASVH Runtime Architecture.

The glossary is intended to provide consistent terminology and ensure that key architectural concepts are interpreted consistently throughout the ASVH document set.

Unless otherwise stated, the definitions provided in this appendix apply only to the ASVH reference architecture.

A.2 Architectural Terms

AI-enabled Ambient Scribing

A software system that captures clinical conversations and generates draft clinical documentation using speech recognition and artificial intelligence technologies.

Authority Receipt

An immutable engineering evidence artefact generated by the Runtime Authority following deterministic runtime evaluation.

The Authority Receipt records the Runtime Context evaluated, Runtime Rule outcomes, Runtime Decision and supporting metadata required for audit, deterministic replay and architectural verification.

The Authority Receipt is distinct from the EPR.

Clinical Accountability

The professional responsibility of the authorised clinician to review, amend where necessary and approve AI-generated clinical documentation prior to execution.

Clinical accountability remains outside the responsibilities of the Runtime Authority.

Deterministic Evaluation

A method of runtime evaluation in which an identical Runtime Context always produces an identical Runtime Decision.

Deterministic evaluation excludes probabilistic reasoning, adaptive optimisation and non-repeatable decision processes.

EPR

The institutional clinical record in which authorised clinical documentation is stored following successful Runtime Authority evaluation.

The EPR is not responsible for Runtime Authority evaluation or Authority Receipt generation.

Execution Bind Point

The final architectural boundary immediately before institutional consequence formation.

At the Execution Bind Point, execution is suspended while the Runtime Authority evaluates whether delegated authority remains legitimately exercisable using the current Runtime Context.

Execution Coordinator

The architectural component responsible for assembling the Runtime Context, invoking the Runtime Authority and acting upon the returned Runtime Decision.

The Execution Coordinator performs no runtime evaluation.

Institutional Consequence

The point at which an approved action creates an operational, organisational or clinical outcome recognised by the institution.

Within ASVH, institutional consequence occurs when authorised clinical documentation is committed to the EPR.

Responsible Clinician

The authorised healthcare professional responsible for reviewing and approving AI-generated clinical documentation before execution.

The Runtime Authority neither replaces nor influences clinician judgement.

Runtime Authority

An independent architectural component responsible for determining whether delegated authority remains legitimately exercisable immediately before institutional consequence formation.

The Runtime Authority evaluates Runtime Context using deterministic Runtime Rules and returns one of three Runtime Decisions:

- ALLOW
 - ESCALATE
 - REFUSE
-

Runtime Context

The complete, immutable collection of operational information evaluated by the Runtime Authority immediately before execution.

The Runtime Context represents the current operational state rather than historical workflow information.

Runtime Decision

The deterministic execution outcome returned by the Runtime Authority following evaluation of the Runtime Context.

Possible Runtime Decisions are:

- **ALLOW** – execution remains admissible.
 - **ESCALATE** – authorised intervention is required before execution.
 - **REFUSE** – execution is not currently admissible.
-

Runtime Rule

A deterministic engineering rule that evaluates a single runtime condition using information contained within the Runtime Context.

The combined evaluation of all applicable Runtime Rules determines the Runtime Decision.

Runtime Rule Engine

The architectural component responsible for evaluating Runtime Rules during runtime execution.

The Runtime Rule Engine performs no probabilistic reasoning or adaptive optimisation.

Runtime Verification

The deterministic evaluation of Runtime Context immediately before institutional consequence formation to determine whether execution remains admissible.

Traceability

The ability to associate every Runtime Decision with the Runtime Context, Runtime Rule evaluations and Authority Receipt from which it was derived.

Traceability supports deterministic replay, audit and engineering verification.

A.3 Abbreviations

Abbreviation	Meaning
AI	Artificial Intelligence
ASVH	Ambient Scribing Verification Harness
DCB0129	Clinical Risk Management – Manufacture of Health IT Systems
DCB0160	Clinical Risk Management – Deployment and Use of Health IT Systems
EPR	Electronic Patient Record
ICS	Integrated Care System
NHS	National Health Service
ORAI	Open Runtime Authority Interface
RA	Runtime Authority
SPEC	Runtime Authority Specification
TRACE	Guidance-to-Implementation Traceability Matrix

A.4 Terminology Consistency

The terminology defined within this appendix is intended to provide a consistent architectural vocabulary across the ASVH document suite.

Where related FlowSignal publications define the same architectural concepts, equivalent meanings should be assumed unless explicitly stated otherwise.

This consistent terminology supports interoperability between the ASVH Methodology, Specification, Traceability Matrix and Runtime Architecture while preserving the implementation-independent principles upon which the ASVH engineering reference project is based.

Appendix B – Related Work and References

B.1 Purpose

The ASVH Runtime Architecture has been developed within the context of established healthcare guidance, clinical safety standards, enterprise architecture principles and AI governance frameworks.

The purpose of this appendix is to acknowledge the existing bodies of work that informed the architectural context of ASVH and to explain how the Runtime Authority architecture relates to them.

ASVH does not seek to replace these frameworks. Instead, it addresses a specific architectural concern not explicitly defined within them: the deterministic evaluation of execution admissibility immediately before institutional consequence formation.

The publications referenced in this appendix provide the governance, clinical safety, architectural and AI assurance context within which the ASVH Runtime Architecture has been developed.

B.2 NHS England – Guidance on the Use of AI-enabled Ambient Scribing Products in Health and Care Settings

Relationship to ASVH

This guidance forms the primary engineering input to the ASVH programme.

The guidance defines the governance expectations, organisational responsibilities and clinical safety considerations associated with the deployment of AI-enabled ambient scribing products within NHS organisations.

ASVH does not reinterpret or extend NHS guidance. Instead, it applies a structured engineering methodology to identify those guidance statements that represent runtime-verifiable operational conditions. These conditions are subsequently transformed into deterministic Runtime Rules evaluated immediately before AI-generated clinical documentation is committed to the EPR.

The guidance therefore provides the policy and operational context from which the ASVH Runtime Architecture is derived.

B.3 DCB0129 – Clinical Risk Management: Manufacture of Health IT Systems

Relationship to ASVH

DCB0129 defines the clinical risk management responsibilities of organisations developing health IT systems.

ASVH assumes that these responsibilities have already been fulfilled during system design, hazard analysis and clinical safety case development.

The Runtime Authority does not replace DCB0129 activities. Instead, it complements them by evaluating whether runtime execution remains admissible using the current operational state immediately before institutional consequence formation.

B.4 DCB0160 – Clinical Risk Management: Deployment and Use of Health IT Systems

Relationship to ASVH

DCB0160 defines the responsibilities of healthcare organisations deploying and operating health IT systems.

The standard focuses on organisational deployment, governance and ongoing operational safety.

ASVH operates after these governance activities have been completed. Runtime Authority provides deterministic execution-time verification within the operational environment established through DCB0160 rather than replacing organisational clinical safety responsibilities.

B.5 NIST AI Risk Management Framework (AI RMF)

Relationship to ASVH

The NIST AI Risk Management Framework provides guidance for the governance and management of AI-related risks throughout the AI lifecycle.

ASVH complements these governance activities by introducing an implementation-independent Runtime Authority responsible for evaluating execution admissibility immediately before institutional consequence formation.

The Runtime Authority therefore represents an execution-time assurance mechanism operating within a broader AI governance framework.

B.6 NIST SP 800-207 – Zero Trust Architecture

Relationship to ASVH

Zero Trust Architecture introduced the principle that trust should be established through continuous verification rather than persistent assumptions.

ASVH applies a similar architectural philosophy to delegated execution authority.

Rather than assuming that previous approvals remain valid, the Runtime Authority independently evaluates whether execution remains admissible using the Runtime Context available immediately before execution.

Unlike Zero Trust, however, ASVH evaluates execution authority rather than identity, network access or device trust.

B.7 Ambient Scribing Platforms

Relationship to ASVH

Existing ambient scribing platforms focus primarily on:

- consultation capture;

- speech recognition;
- transcription;
- AI-assisted documentation generation;
- clinician review.

ASVH intentionally avoids prescribing the internal architecture of these systems.

Instead, the Runtime Authority operates as an independent architectural component positioned after clinician approval and immediately before commitment to the EPR.

This enables the architecture to complement existing suppliers rather than replace them.

B.8 FlowSignal Runtime Authority Research

Relationship to ASVH

The ASVH Runtime Architecture builds upon the broader FlowSignal Runtime Authority research programme.

Previous FlowSignal publications define the principles of implementation-independent Runtime Authority, deterministic runtime evaluation and the Execution Bind Point.

ASVH applies those architectural principles to a specific healthcare implementation derived from NHS England guidance on AI-enabled ambient scribing.

B.9 Contribution of the ASVH Runtime Architecture

The ASVH Runtime Architecture contributes an implementation-independent reference model for deterministic runtime verification within AI-enabled ambient scribing workflows.

Rather than redefining governance, clinical safety or AI generation, the architecture introduces a distinct architectural responsibility:

The independent evaluation of whether delegated authority remains legitimately exercisable immediately before institutional consequence formation.

This contribution complements existing healthcare guidance by demonstrating how runtime-verifiable operational conditions can be transformed into deterministic Runtime Rules evaluated at the Execution Bind Point.

B.10 References

1. NHS England. *Guidance on the Use of AI-enabled Ambient Scribing Products in Health and Care Settings*.
2. NHS England. *Ambient Voice Technology (AVT) Supplier Registry*.
3. NHS Digital. *DCB0129: Clinical Risk Management – Its Application in the Manufacture of Health IT Systems*.
4. NHS Digital. *DCB0160: Clinical Risk Management – Its Application in the Deployment and Use of Health IT Systems*.
5. National Institute of Standards and Technology (NIST). *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*.
6. National Institute of Standards and Technology (NIST). *SP 800-207: Zero Trust Architecture*.
7. FlowSignal™. *ASVH-METH-001 – Runtime Requirements Engineering Methodology*.
8. FlowSignal™. *ASVH-SPEC-001 – Runtime Authority Specification*.
9. FlowSignal™. *ASVH-TRACE-001 – Guidance-to-Implementation Traceability Matrix*.
10. FlowSignal™. *ORAI-001 – Open Runtime Authority Interface*.
11. FlowSignal™. *FEAL-001 – Institutional Runtime Architecture*.
12. FlowSignal™. *IRAI-AN-003 – Industry Convergence Around Runtime Authority*.

Appendix C – Architectural Design Decisions

C.1 Purpose

This appendix records the principal architectural design decisions made during the development of the ASVH Runtime Architecture.

The objective is to provide transparency regarding the rationale behind significant architectural choices, supporting independent review, future evolution and implementation consistency.

Each design decision records the architectural problem addressed, the adopted solution and the rationale supporting that decision.

C.2 Architectural Decision Register

Decision ID	Design Decision	Rationale
ADR-001	Runtime Authority positioned after clinician approval	Preserves clinical accountability while ensuring execution is evaluated immediately before institutional consequence formation.
ADR-002	Runtime Authority independent of ambient scribing platform	Prevents coupling between AI generation and execution assurance, enabling supplier independence.
ADR-003	Runtime Authority positioned before EPR commitment	Ensures execution may still be prevented before institutional consequence occurs.

ADR-004	Runtime Context assembled externally	Separates evidence collection from evidence evaluation, preserving Runtime Authority independence.
ADR-005	Runtime Context immutable during evaluation	Supports deterministic replay, audit and reproducibility.
ADR-006	Deterministic Runtime Rules	Ensures identical Runtime Context produces identical Runtime Decisions.
ADR-007	Three Runtime Decisions (ALLOW, ESCALATE, REFUSE)	Provides clear and deterministic execution outcomes while supporting operational intervention where appropriate.
ADR-008	Authority Receipt generated independently	Separates execution evidence from execution outcome and supports evidential accountability.
ADR-009	Authority Receipt not part of the EPR	Maintains separation between engineering evidence and the institutional clinical record.
ADR-010	Implementation-independent architecture	Enables Runtime Authority to integrate with multiple ambient scribing suppliers and healthcare environments.

C.3 Architectural Decision Principles

Each architectural decision recorded within this appendix has been evaluated against the following principles:

- Independence
- Determinism
- Separation of Responsibility
- Implementation Independence
- Evidential Accountability
- Traceability
- Replayability
- Clinical Accountability

Architectural changes should preserve these principles unless explicitly justified through future revisions of the ASVH reference architecture.

C.4 Alternative Approaches Considered

During the development of the ASVH Runtime Architecture, several alternative architectural approaches were considered.

Runtime Authority before clinician approval

Rejected

Would require evaluation of documentation that has not yet been accepted by the responsible clinician, weakening the separation between clinical judgement and runtime verification.

Runtime Authority integrated within the ambient scribing platform

Rejected

Would tightly couple runtime verification to supplier-specific implementations, reducing portability and implementation independence.

Runtime Authority after EPR commitment

Rejected

Would reduce runtime verification to retrospective audit after institutional consequence has already occurred.

Mutable Runtime Context

Rejected

Would compromise deterministic replay and reduce confidence in evidential integrity.

Runtime Decisions without Authority Receipts

Rejected

Would provide an operational outcome but insufficient engineering evidence to support audit, traceability or deterministic replay.

C.5 Architectural Constraints

Every implementation claiming conformance with the ASVH Runtime Architecture should preserve the following architectural constraints:

- Runtime Authority remains independent.
- Runtime evaluation remains deterministic.
- Runtime Context remains immutable during evaluation.
- Runtime Rules evaluate only Runtime Context.
- Authority Receipts remain complete and traceable.
- Execution occurs only following an **ALLOW** decision.
- Institutional consequence occurs only after Runtime Authority evaluation.

These constraints define the architectural characteristics of ASVH independently of implementation technology.

C.6 Future Evolution

The architectural decisions recorded within this appendix represent the design baseline for Version 1.0 of the ASVH Runtime Architecture.

Future revisions may introduce additional architectural capabilities provided that they do not compromise the core principles of:

- implementation independence;
- deterministic evaluation;
- execution-time verification;
- evidential accountability; and
- separation of architectural responsibility.

Where architectural decisions change in future versions, the corresponding decision record should be updated to preserve the historical evolution of the reference architecture.

Appendix D – ASVH Document Relationship and Architectural Traceability

D.1 Purpose

The ASVH engineering reference project comprises a collection of complementary documents, each addressing a distinct aspect of deterministic runtime verification for AI-enabled ambient scribing.

This appendix describes the relationship between those documents and explains how they collectively define the ASVH engineering reference architecture.

The purpose is to demonstrate that the ASVH document set forms a coherent architectural framework rather than a collection of independent publications.

D.2 ASVH Engineering Document Set

Document	Primary Purpose
ASVH-METH-001	Defines the engineering methodology used to derive runtime requirements from NHS England guidance.
ASVH-SPEC-001	Specifies the deterministic Runtime Rules and Runtime Authority behaviour.
ASVH-TRACE-001	Demonstrates traceability between NHS England guidance, runtime requirements and Runtime Rules.
ASVH-ARCH-001	Defines the implementation-independent reference architecture supporting Runtime Authority.

Together, these documents define the engineering lifecycle from published guidance through to architectural implementation.

D.3 Engineering Lifecycle

The ASVH document set follows a structured engineering progression.

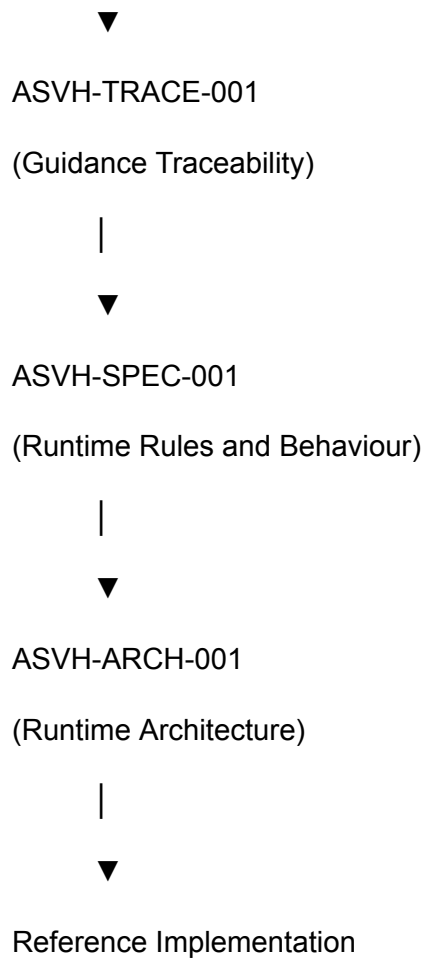
NHS England Guidance



ASVH-METH-001

(Runtime Requirements Engineering)





Each document builds upon the outputs of the preceding document while maintaining clear separation of engineering responsibilities.

D.4 Architectural Traceability

The ASVH Runtime Architecture is fully traceable to published NHS England guidance through the supporting ASVH engineering documents.

The relationship between documents can be summarised as follows.

Architectural Element	Supporting Document
Runtime Requirements	ASVH-METH-001

Guidance Traceability	ASVH-TRACE-001
Runtime Rules	ASVH-SPEC-001
Runtime Context	ASVH-SPEC-001 / ASVH-ARCH-001
Runtime Authority	ASVH-ARCH-001
Authority Receipt	ASVH-ARCH-001 / ASVH-SPEC-001
Reference Implementation	ASVH-001

This traceability demonstrates that architectural decisions are derived from documented engineering artefacts rather than implementation-specific assumptions.

D.5 Relationship to the Reference Implementation

The ASVH Runtime Architecture defines the implementation-independent architectural model.

The accompanying reference implementation demonstrates one practical implementation of that architecture.

The implementation is intended to validate the feasibility of the architectural principles defined within this document and does not constrain future implementations.

Alternative implementations may adopt different technologies, deployment models or integration approaches provided that they preserve the architectural principles established by the ASVH Runtime Architecture.

D.6 Relationship to the FlowSignal Architecture

The ASVH Runtime Architecture forms part of the broader FlowSignal Runtime Authority research programme.

Within this programme:

- **FEAL-001** defines the Institutional Runtime Architecture.
- **ORAI-001** defines implementation-independent interoperability principles.
- **ASVH** demonstrates a healthcare-specific implementation derived from NHS England guidance.

The ASVH architecture therefore represents one domain-specific application of the broader Runtime Authority architectural model.

D.7 Future Evolution

The ASVH engineering reference project is intended to evolve as healthcare guidance, implementation experience and interoperability requirements mature.

Future revisions may extend the architecture through:

- additional runtime verification scenarios;
- expanded Runtime Rule libraries;
- enhanced interoperability profiles;
- new healthcare implementation examples;
- alignment with future NHS guidance and industry standards.

Such enhancements should preserve the implementation-independent principles established throughout the ASVH document suite.

D.8 Appendix Summary

This appendix demonstrates that the ASVH engineering publications form a coherent and traceable architectural framework.

Each document addresses a distinct engineering concern while contributing to a common objective: the implementation-independent, deterministic evaluation of runtime admissibility immediately before institutional consequence formation.

By maintaining clear separation between methodology, traceability, specification, architecture and implementation, the ASVH reference project provides a structured engineering foundation that can be reviewed, implemented and evolved without compromising the core architectural principles established by the Runtime Authority.

Document Classification

Engineering Reference Architecture

Document Identifier

ASVH-ARCH-001

Version

Version 1.0

Status

Public Release

Related Publications

- ASVH-METH-001
- ASVH-SPEC-001
- ASVH-TRACE-001
- ASVH-001 (Reference Implementation Report)